# Maximum Blackouts That Keep Every Lattice Rectangle Identifiable

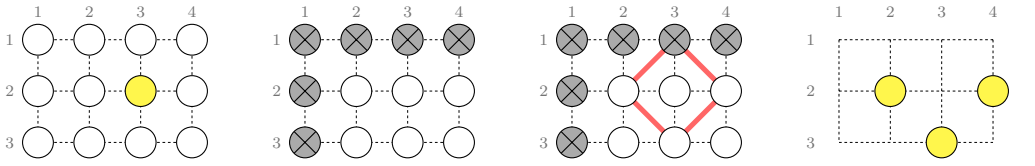
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**Abstract**

Kagey's Problem 001 concerns a grid of points, some of them blacked out. A rectangle with its corners on grid points, tilted or not, shows only its white corners. We ask how many points can be blacked out while every rectangle stays determined by what it shows. We study the largest such blackout and the number of blackouts of that size, the tables A397315 and A399580 of the On-Line Encyclopedia of Integer Sequences. A graph on the rows and columns and Mantel's theorem bound the maximum by the width plus a quarter of the squared height. We find the maximum on up to five rows at every width, and on six rows from width 15 on. On five rows from width eight on, and on the grid with six rows and eight columns, the maximum exceeds the sum of the two sides, against a guess we recorded there. On three rows we count the optima exactly, and on four rows we find their growth rate. For every fixed number of rows the maximum minus the width is eventually periodic. Some values rest on computer checks.

## 1 Introduction

You are shown a grid of points with $h$ rows and $m$ columns, and you black out some of them. We call the set of blacked-out points the *blackout*. Then



(a) the point $(3, 2)$ (b) our blackout (c) the rectangle (d) what we are shown

Figure 1: One round on the $3 \times 4$ grid. Gray crossed circles are blacked out and white circles are visible. In (c) the red outline is the rectangle the game master draws, and in (d) the yellow circles are the corners we are shown.

the game master draws a rectangle with its four corners on grid points, tilted or not. You are told which of its corners are white, and you must say where all four corners are. Kagey asks how many points you can black out so that no two rectangles ever look the same.

We write $(i, j)$ for the point in column $i$ and row $j$, counting columns from the left and rows from the top. Figure 1 plays one round on the $3 \times 4$ grid with row 1 and column 1 blacked out. The game master draws the tilted square with corners $(2, 2)$, $(3, 1)$, $(4, 2)$, and $(3, 3)$, and we are shown $(2, 2)$, $(4, 2)$, and $(3, 3)$. Three corners of a rectangle determine the fourth (Lemma 4), so the fourth corner is $(3, 1)$. This blackout of six points is good, and no good blackout of the $3 \times 4$ grid is larger (Section 5).

Kagey poses the problem in his Open Problem Collection [24] and in his gift for the fourteenth Gathering 4 Gardner [23]. Kagey's page for the problem cites his earlier question [22] about upright triangles in a triangular array, where Lavrov gave a counting upper bound and an alteration lower bound. We write $f(h, m)$ for the largest size of a good blackout of the $h \times m$ grid, and $a_h(m)$ for the number of good blackouts of that size, counted with their coordinates. From Section 2 on, a good blackout is called *valid* (Definition 1). These are the tables A397315 and A399580 of the OEIS [37], which we contributed. The first versions of these entries recorded the conjecture $f(3, m) = m+2$ and the guess that $f(h, m) = h+m$ whenever $\min(h, m) \geq 4$, apart from $f(4, 4) = 7$ and $f(4, 5) = 8$. We show that the guess fails. The three-row conjecture was proved in Lean by The Omega Institute [33]. It is also the case $h = 3$ of our row bound, and we do not claim it as new.

**Results.** Two rectangles look the same exactly when every corner that belongs to only one of them is black (Lemma 2). Put $q_h = \lfloor h^2/4 \rfloor$. Computer

checks are marked in each statement.

**Theorem A** (narrow grids). *(a) For all $h, m \geq 1$ we have $f(h,m) \geq h + m - 1$, and for $m \geq 3$ we have $f(h,m) \leq m + q_h$.*

*(b) For $m \geq 3$ we have $f(2,m) = m+1$ and $f(3,m) = m+2$. Also $f(4,m) = m+4$ for every $m \geq 6$, and $f(5,5) = 10$.*

*(c) (Computer-assisted.) We have $f(4,4) = 7$, $f(4,5) = 8$, $f(5,6) = 11$, and $f(5,7) = 12$, while $f(5,m) = m+6$ for every $m \geq 8$.*

*(d) (Computer-assisted for $m \geq 9$.) For $8 \leq m \leq 14$ we have $f(6,m) \leq m+8$, and $f(6,m) = m+9$ for every $m \geq 15$.*

*(e) (Computer-assisted.) We have $f(6,6) = 12$, $f(6,7) = 13$, $f(7,7) = 14$, and $f(6,8) = 15$.*

Part (a) is Lemma 16 and Theorem 8. So for $h \leq 6$ the row bound is attained at every large width, and the guess $f(h,m) = h + m$ first fails on the $5 \times 8$ grid (Theorem 44).

**Theorem B** (three-row optima). *For $m \geq 4$ we have $a_3(m) = 4(m-1)(3m+2)\,3^{m-5} + E_m$, where $E_m > 0$ counts the optima with a column black in rows 1 and 3, and $E_m = O(m^2(1+\sqrt{3})^m)$. So $a_3(m) \sim \frac{4}{81} m^2 3^m$, and a fraction tending to $4/27$ of the $m^2 3^{m-1}$ axis-valid candidates is valid. The series $\sum_{m \geq 3} a_3(m) z^m$ is rational. (Computer-assisted.) Its reduced denominator is $(1-3z)^3(1-2z-2z^2)^3(1+2z^2)^3$, and $E_m$ has a closed form with $E_m \sim \frac{19+11\sqrt{3}}{432} m^2 (1+\sqrt{3})^m$.*

These are Theorems 24, 27, and 25, the last finished in Section 10.

**Theorem C** (four rows). *Let $m \geq 4$. No valid blackout of the $4 \times m$ grid hides all four corners of a rectangle. For $m \geq 6$,*

$$a_4(m) \leq 3\,(m)_4\, 4^{m-4} \Big(\frac{53}{64}\Big)^{\max(0,\lfloor m/4 \rfloor - 4)}, \qquad (m)_4 = m(m-1)(m-2)(m-3).$$

*(Computer-assisted.) The series $\sum_{m \geq 6} a_4(m) z^m$ is rational with reduced denominator of degree 420, and $a_4(m) \sim c\, m^4 \rho_4^m$ with $c \approx 8.1611 \times 10^{-5}$ and $\rho_4 = 3.35199\ldots$, an algebraic number of degree 20.*

The base $212^{1/4} = 3.8157\ldots < 4$ makes valid blackouts rare (Theorems 32, 34, 37).

**Theorem D** (finite collar)**.** *Let $h \geq 3$ and $W_h = 1 + \lfloor (h-1)^2/4 \rfloor$. Pad a blackout of an $h \times m$ grid, on one side or on both, by columns black only in the top row, or only in the bottom row. Then every such padding is valid if and only if the padding by $W_h$ columns on each chosen side is valid. (Computer-assisted.) A blackout of the $5 \times 8$ grid and one of the $6 \times 16$ grid pass this test with the top row on the right. Hence $f(5, m) = m + 6$ for $m \geq 8$ and $f(6, m) = m + 9$ for $m \geq 16$, and a stored blackout of the $6 \times 15$ grid gives $f(6, 15) = 24$.*

See Theorem 39 and Section 7. The padding row must be a boundary row (Remark 40).

**Theorem E** (every fixed height)**.** *Fix $h \geq 1$ and write a blackout of the $h \times m$ grid as the word $B_1 \cdots B_m$, where $B_i$ is the set of black rows of column $i$. The valid words form a regular language. If $h \geq 2$, the excess $f(h, m) - m$ is eventually periodic in $m$, and $\sum_{m \geq 0} a_h(m) z^m$ is rational.*

These are Theorems 48 and 49, and they give no period and no threshold. On square grids, Section 11 shows that $f(n, n) \geq c\, n^{6/5} (1 + \ln n)^{-2/5}$ for some $c > 0$ and all large $n$.

**Prior work.** In its hitting-set form (Corollary 3) the game asks for a test cover [25, 31, 5, 16]. For any $n$ distinct subsets of an $n$-set, Bondy's theorem [7] finds a point whose deletion keeps them distinct, but here the rectangles far outnumber the points. A fully black axis-aligned rectangle is a four-cycle in a bipartite graph on the rows and columns (Section 4), so forbidding these alone is the problem of Zarankiewicz [35]. The axis-aligned rectangles alone lead to bipartite graphs without cycles of length four and six (Lemma 7). Such graphs have $O(n^{4/3})$ edges on $n + n$ vertices [8]. They appear in a question of Erdős (Problem 1080 of Bloom's list [6]), and de Caen and Székely [10, 11] study the unbalanced case. Neuwirth [32, Theorem 1(ii) and Proposition 17] bounds such graphs on $h + m$ vertices by $m + h^2/4$ when $4m \geq h^2$, and by a bound that is no larger for smaller $m$. So the row bound is known in graph form. Our proof with Mantel's theorem [27] also finds and counts the equality case. None of these works sees tilted rectangles. Grids of fixed height are often treated by column transfer [13, 28, 1], [14, Proposition V.6]. Klavžar and Žerovnik [26] compute domination numbers of fixed-height grids with $(\min, +)$ transfer matrices, those of all grids are known [17], and minimum identifying codes of circular and infinite grids of

small height were found by computer search [9]. Jiang [21] proves that the least density of an identifying code in an infinite strip is attained by a periodic code, and Theorem E is of the same kind. Given the regularity in Theorem 48, repeated use of Theorem 1(3) of Guilhot, Little, and Parkinson [18] also implies the periodicity and the rational series (Section 8). We prove them directly.

## 2 Conventions and the difference lemma

These conventions hold throughout the paper.

(1) An $h \times m$ grid has $h$ rows and $m$ columns, and $V$ is its set of points. A $2 \times m$ grid is a *strip*, and an $n \times n$ grid is a *square*.

(2) The point $(i, j)$ is in column $i$ and row $j$, and $(1, 1)$ is the top-left point. So $i$ grows to the right and $j$ grows downward, as in Figure 1(a).

(3) A *rectangle* is a set of four grid points that are the corners of a rectangle of positive area, and tilted rectangles count. We identify a rectangle $R$ with its corner set $C(R)$. It is *axis-aligned* if its sides are parallel to the rows and columns, and *tilted* otherwise. Four distinct points form a rectangle if and only if they split into two segments with the same midpoint and the same length.

(4) A *blackout* is a set $S \subseteq V$, and $K = V \setminus S$ is the *kept set*. The *presentation* of a rectangle $R$ is the set $P_S(R) = C(R) \setminus S$ of its white corners.

(5) The *black rows* of column $i$ form the set $B_i$ of rows $j$ with $(i, j) \in S$. Column $i$ is *empty*, a *singleton*, a *double*, or a *triple* when $|B_i|$ is 0, 1, 2, or 3. It is *full* when $|B_i| = h$.

(6) The *pair set* $F$ is the set of row pairs contained in some $B_i$. We put $q_h = \lfloor h^2/4 \rfloor$, the quarter-square A002620$(h)$.

**Definition 1** (valid blackout)**.** A blackout $S$ is *valid* if different rectangles always have different presentations, that is, if $P_S(R_1) \neq P_S(R_2)$ whenever $R_1 \neq R_2$. The *maximum blackout* $f(h, m)$ is the largest size of a valid blackout of the $h \times m$ grid. We write $a_h(m)$ for the number of valid blackouts of size $f(h, m)$. They are counted with their coordinates, not up to symmetry.

Swapping rows and columns gives $f(h,m) = f(m,h)$ and $a_h(m) = a_m(h)$. The tables $f(h,m)$ and $a_h(m)$ are A397315 and A399580, read by antidiagonals with both indices starting at 2. Section 8 also uses width 0 and height 1, with no rectangles, where $f(h,0) = 0$, $a_h(0) = 1$, $f(1,m) = m$, and $a_1(m) = 1$.

The empty presentation is allowed. It means that every corner is black, so a valid blackout hides at most one rectangle.

**Lemma 2** (difference sets)**.** *For all sets $A$, $B$, and $S$ we have $A \setminus S = B \setminus S$ if and only if $A \triangle B \subseteq S$, where $\triangle$ is the symmetric difference.*

*Proof.* Both conditions say that no point outside $S$ lies in exactly one of $A$ and $B$. □

**Corollary 3** (hitting sets)**.** *A blackout $S$ is valid if and only if the kept set $K$ meets $C(R_1) \triangle C(R_2)$ for every pair of distinct rectangles $R_1$, $R_2$. So $hm - f(h,m)$ is the least size of a set meeting every such difference set.*

*Proof.* Apply Lemma 2 to $C(R_1)$, $C(R_2)$, and $S$ for every pair. □

A larger kept set meets more sets, so every subset of a valid blackout is valid. Hence a search that finds no valid blackout of exactly $k$ points proves $f(h,m) < k$.

**Lemma 4** (three corners suffice)**.** *Three corners of a rectangle determine the rectangle. So two distinct rectangles with the same presentation both show at most two corners.*

*Proof.* Three corners $p$, $q$, and $r$ of a rectangle form a right triangle, with the right angle at the corner adjacent to both others, say $q$. A triangle has only one right angle, so the three points determine $q$ and the fourth corner $p + r - q$. □

# 3 Strips

On two rows every rectangle is axis-aligned. Indeed, a tilted rectangle has no horizontal side, so both sides at a lowest corner go strictly up, and the opposite corner lies in a third row.

**Theorem 5** (strip theorem)**.** *Let $m \geq 3$.*

(a) *A blackout of the* $2 \times m$ *strip is valid if and only if at most one of its columns is full.*

(b) *We have* $f(2, m) = m + 1$*, and the maximum blackouts are those with one full column and one black point in every other column.*

(c) *There are* $a_2(m) = m \cdot 2^{m-1}$ *maximum blackouts.*

*For* $m = 2$ *there is one rectangle, so* $f(2, 2) = 4$ *and* $a_2(2) = 1$*.*

*Proof.* (a) Two distinct rectangles share at most one column. If they share one, it cancels, and the difference set is the four points of the other two columns $p$ and $q$, the *pair constraint* on $\{p, q\}$. If they share no column, the difference set is the eight points of four columns, and it contains a pair constraint. Every pair constraint occurs, since $m \geq 3$ gives a third column $z$, and the rectangles on $\{z, p\}$ and $\{z, q\}$ have exactly this difference set. By Corollary 3, $S$ is valid if and only if $K$ meets every two columns, that is, if and only if at most one column has no kept point.

(b) By (a) a valid $S$ has $|K| \geq m - 1$, so $|S| \leq m + 1$. Equality holds exactly when one column is full and every other column has one kept point, hence one black point. Such blackouts are valid by (a).

(c) There are $m$ choices for the full column and two in each other column. □

So row 2 of A399580 is A001787$(m) = m2^{m-1}$ from $m = 3$ on.

**Proposition 6** (orbits and kings)**.** *For* $m \geq 3$ *the number* $o_m$ *of maximum blackouts of the* $2 \times m$ *strip up to the row reflection and the column reversal is* $m\, 2^{m-3} + [m \text{ odd}]\, 2^{(m-3)/2}$*. It equals* A322284$(m - 1)$*, the number of ways to place* $m - 1$ *nonattacking kings on a* $2 \times (2m - 2)$ *board up to the symmetries of the rectangle. So* $o_3, \ldots, o_8$ *are* $4, 8, 22, 48, 116, 256$*.*

*Proof.* By Burnside's lemma $4o_m$ is $a_2(m)$ plus the numbers of optima fixed by the reflection, which is 0 since it moves every singleton column, and by the reversal and the half turn. These fix none for even $m$, and $2^{(m-1)/2}$ each for odd $m$, with a central full column.

Put $n = m - 1$. An optimum is given by its full column $e$ and the rows $r_1, \ldots, r_n$ of its other black points, from left to right. Two kings attack exactly when their columns are equal or adjacent. So $n$ nonattacking kings on the $2 \times 2n$ board are given by columns $c_1 < \cdots < c_n$ with $c_{j+1} \geq c_j + 2$

and any rows $r_1, \ldots, r_n$. The $n+1$ gaps $c_1 - 1$, $c_{j+1} - c_j - 2$ for $1 \le j < n$, and $2n - c_n$ are nonnegative and sum to 1. So exactly one of them is 1, say the $e$th, and $c_j = 2j - 1 + [j \ge e]$. Hence the kings at $(c_j, r_j)$ correspond bijectively to the optima. The reflection swaps the rows in both pictures. The reversal sends $e$ to $n + 2 - e$ and reverses the rows, and on the board it sends $c$ to $2n + 1 - c$, where $2n + 1 - c_{n+1-j} = 2j - 1 + [j \ge n + 2 - e]$. So the bijection commutes with both actions, and the orbits correspond. □

# 4 The row-column graph and general bounds

This section proves $h + m - 1 \le f(h, m)$, and $f(h, m) \le m + q_h$ for $m \ge 3$, and lists the tilted rectangles of narrow grids. A blackout $S$ of the $h \times m$ grid is *axis-valid* if no two axis-aligned rectangles have the same presentation. A rectangle that shows at most two corners is *weak*, and by Lemma 4 only weak rectangles can be confused.

The *row-column graph* $G_S$ has a vertex $r_j$ for each row $j$, a vertex $c_i$ for each column $i$, and an edge $r_j c_i$ for each black point $(i, j)$. A fully black axis-aligned rectangle on rows $j, j'$ and columns $i, i'$ is the cycle $r_j\, c_i\, r_{j'}\, c_{i'}$.

**Lemma 7** (axis-aligned rectangles and short cycles)**.** *If $G_S$ has no four-cycle and no six-cycle, then $S$ is axis-valid. If $m \ge 3$ and $S$ is axis-valid, then $G_S$ has no four-cycle and no six-cycle.*

*Proof.* Let $m \ge 3$. A four-cycle $r_j\, c_i\, r_{j'}\, c_{i'}$ and a third column $w$ give the rectangles on rows $\{j, j'\}$ and columns $\{i, w\}$ or $\{i', w\}$, which both show the white points among $(w, j)$ and $(w, j')$. For a six-cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ the different rectangles on rows $\{a, b\}$, columns $\{x, y\}$ and on rows $\{a, c\}$, columns $\{y, z\}$ are black except possibly at $(y, a)$.

Conversely, let $R_1 \ne R_2$ be axis-aligned with the same presentation, so $C(R_1) \triangle C(R_2) \subseteq S$ by Lemma 2. Opposite corners determine an axis-aligned rectangle, so the common corners lie in one row or one column. Two common corners $(x, a)$ and $(y, a)$ leave rows $b \ne c$ and the black four-cycle $r_b\, c_x\, r_c\, c_y$, and a common column is similar. If $(x, a)$ is the only common corner, with $R_1$ on rows $\{a, b\}$ and columns $\{x, y\}$ and $R_2$ on rows $\{a, c\}$ and columns $\{x, z\}$, then $r_a\, c_y\, r_b\, c_x\, r_c\, c_z$ is a black six-cycle. With no common corner, $R_1$ is a black four-cycle. The all-black $2 \times 2$ grid shows that $m \ge 3$ is needed. □

**Theorem 8** (row bound)**.** *Let $m \geq 3$. Every axis-valid blackout of the $h \times m$ grid has at most $m + q_h$ points, and at most $h + q_m$ points if also $h \geq 3$. Hence $f(h, m) \leq m + q_h$ for $m \geq 3$.*

For each nonempty column $i$ choose a tree $T_i$ with vertex set $B_i$, and let $H$ be the graph on the rows whose edges are those of all the trees $T_i$.

**Lemma 9** (row graph)**.** *Let $m \geq 3$ and let $S$ be axis-valid, with arbitrary trees $T_i$. Then $H$ has no repeated edge and no triangle, and $e(H) = |S| - t$, where $t$ is the number of nonempty columns.*

*Proof.* By Lemma 7, $G_S$ has no four-cycle and no six-cycle. If columns $i \neq i'$ both gave the edge $r_j r_{j'}$, then $r_j\, c_i\, r_{j'}\, c_{i'}$ would be a four-cycle. So $e(H) = \sum(|B_i| - 1) = |S| - t$. A triangle $r_a r_b r_c$ does not lie in one tree. If two of its edges come from column $i$ and one from column $i'$, then two of $a, b, c \in B_i$ lie in $B_{i'}$, a four-cycle. If its edges come from columns $i, i', i''$, then $r_a\, c_i\, r_b\, c_{i'}\, r_c\, c_{i''}$ is a six-cycle. □

**Lemma 10** (Mantel [27])**.** *Let $H$ be a simple triangle-free graph on $h$ vertices, with degrees $d(\cdot)$. Let $v$ have maximum degree $\Delta$, let $A$ be its set of neighbors, and let $B$ be the set of the other $h - \Delta$ vertices, with $e(B)$ edges inside $B$. Then*

$$e(H) = \sum_{b \in B} d(b) - e(B) \leq (h - \Delta)\Delta - e(B) \leq q_h.$$

*Proof.* No two vertices of $A$ are adjacent, so every edge has an end in $B$, and the degree sum over $B$ counts the edges inside $B$ twice. Every degree is at most $\Delta$, $e(B) \geq 0$, and $(h - \Delta)\Delta$ is an integer at most $h^2/4$. □

*Proof of Theorem 8.* Lemmas 9 and 10 give $|S| = t + e(H) \leq m + q_h$. Only Lemma 7 uses a third column, so for $h \geq 3$ the exchange of rows and columns gives $|S| \leq h + q_m$. □

Neuwirth [32, Theorem 1(ii)] proves the graph form when $4m \geq h^2$, and for smaller $m$ his Proposition 17 gives a bound that is at least as strong. His construction showing that the bound is sharp [32, Proposition 17] gives blackouts of the next theorem, which adds the converse and the count.

**Theorem 11** (equality structure)**.** *Let $h \geq 4$ and $m \geq 3$, and let $S$ be a blackout of the $h \times m$ grid with $m + q_h$ points. Then $S$ is axis-valid if and only if*

*(a) every column is a singleton or a double,*

*(b) exactly $q_h$ columns are doubles, with pairwise distinct row pairs, and*

*(c) these row pairs are the edges of a complete bipartite graph on the rows whose parts differ in size by at most one.*

*There are $p_h\,(m)_{q_h}\,h^{m-q_h}$ such blackouts, where $(m)_q = m(m-1)\cdots(m-q+1)$ and $p_h$, the number of such unordered partitions of the rows, is $\frac{1}{2}\binom{h}{h/2}$ for even $h$ and $\binom{h}{\lfloor h/2\rfloor}$ for odd $h$.*

*Proof. Step 1. Equality in Mantel's theorem.* Let $S$ be axis-valid and build $H$ from arbitrary trees. Since $|S| = m + q_h$, Lemmas 9 and 10 force $t = m$ and equality throughout Lemma 10. So every column is nonempty, $e(B) = 0$, and every $b \in B$ has its $\Delta = |A|$ neighbors in $A$. Hence $H$ is complete bipartite with parts $A$ and $B$, whose sizes differ by at most one since $(h-\Delta)\Delta = q_h$.

*Step 2. No triple.* Suppose $|B_i| \geq 3$, so $T_i$ has a vertex $v$ with two neighbors $u, w$, both in the part opposite to $v$. Since $h \geq 4$ the part of $v$ has a vertex $z \neq v$. Replacing the edge $vw$ of $T_i$ by $uw$ gives a tree on $B_i$, and the new graph $H'$ keeps $uz$ and $wz$ since $z \neq v$. The triangle $uwz$ contradicts Lemma 9 and proves (a).

*Step 3. The doubles.* By (a) the edges of $H$ are the row pairs of the $e(H) = q_h$ doubles. Lemma 9 gives (b) and Step 1 gives (c).

*Step 4. The converse.* Under (a) and (b) no two columns share two black rows, and a six-cycle $r_a\,c_x\,r_b\,c_y\,r_c\,c_z$ would need doubles $\{a,b\}$, $\{b,c\}$, $\{a,c\}$, a triangle in the graph of (c). So Lemma 7 applies.

*Step 5. The count.* Choose a partition, an injective assignment of its $q_h$ edges to columns, and a row for each other column. The doubles determine these choices. □

**Corollary 12** (small widths)**.** *Let $h \geq 4$ and $3 \leq m < q_h$. Then $f(h,m) \leq m + q_h - 1$. In particular $f(5,5) \leq 10$.*

*Proof.* Use Theorem 8. A valid blackout of $m + q_h$ points is axis-valid, so by Theorem 11(b) it has $q_h \leq m$ doubles. Note that $q_5 = 6$. □

Neuwirth's Proposition 17 [32] also gives this bound in graph form.

For tilted rectangles put $W_h = 1 + \lfloor (h-1)^2/4 \rfloor$, so $W_3 = 2$, $W_4 = 3$, and $W_5 = 5$.

**Lemma 13** (leftmost corner)**.** *A tilted rectangle $R$ has exactly one corner of smallest column and exactly one of smallest row. If $\ell = (x, r)$ is the leftmost corner, the sides from $\ell$ are $(a, -b)$ and $(c, d)$ with positive integers $a, b, c, d$ and $ac = bd$, and the corners are*

$$(x, r), \qquad (x + a,\, r - b), \qquad (x + c,\, r + d), \qquad (x + a + c,\, r - b + d). \quad (1)$$

*So $R$ occupies the rows $r - b$ to $r + d$, its topmost corner is $(x + a, r - b)$, and its column span is $a + c$.*

*Proof.* No side is vertical, so two corners in the smallest column $x$ are opposite and the center is in column $x$. Then the other two corners, with columns at least $x$ and mean $x$, also lie in column $x$, which is impossible. Rows are similar. So both sides from $\ell$ go strictly to the right, and since they are perpendicular, one goes up and one goes down, with $ac = bd$. □

**Lemma 14** (span bound)**.** *Let $h \geq 3$. Every tilted rectangle of the $h \times m$ grid has span $a + c \leq W_h$, so it meets at most $W_h + 1$ columns. Grids with one or two rows have no tilted rectangles.*

*Proof.* Here $b + d \leq h - 1$, and $(a - 1)(c - 1) \geq 0$ gives $a + c \leq bd + 1 \leq W_h$, since $bd$ is an integer at most $(b + d)^2/4$. With at most two rows, $b + d \leq 1$ is impossible. □

So $W_h$ is A033638$(h-1)$. The sides $(1, -b)$ and $(bd, d)$ with $b = \lfloor (h-1)/2 \rfloor$ and $d = \lceil (h - 1)/2 \rceil$ give a tilted rectangle of span $W_h$, so the bound is attained when $m > W_h$. The same argument with $b + d = n$ proves the conjecture in A338885 that the largest entry in row $n$ there is A033638$(n)$.

For $1 \leq i \leq m - 2$ and $1 \leq j \leq h - 2$ the *diamond* $D(i, j)$ is the square with corners $(i, j + 1)$, $(i + 1, j)$, $(i + 2, j + 1)$, and $(i + 1, j + 2)$. Four rows allow four more kinds, $\mathsf{A}_i$, $\mathsf{B}_i$, $\mathsf{C}_i$, and $\mathsf{E}_i$ for $1 \leq i \leq m - 3$. Each has one corner in each of the columns $i$ to $i + 3$. Read from left to right, the rows of these corners are 3412 for $\mathsf{A}_i$, 3142 for $\mathsf{B}_i$, 2413 for $\mathsf{C}_i$, and 2143 for $\mathsf{E}_i$.

**Lemma 15** (tilted rectangles on narrow grids)**.** *The tilted rectangles of the $3 \times m$ grid are the diamonds $D(i, 1)$ with $1 \leq i \leq m - 2$. Those of the $4 \times m$ grid are the diamonds $D(i, 1)$ and $D(i, 2)$ with $1 \leq i \leq m - 2$ and the rectangles $\mathsf{A}_i, \mathsf{B}_i, \mathsf{C}_i, \mathsf{E}_i$ with $1 \leq i \leq m - 3$.*

*Proof.* Each listed set is a rectangle. On three rows $b = d = 1$, so $a = c = 1$ and $\ell$ is in row 2. On four rows either $b = d = 1$, which gives the diamonds, or $\{b, d\} = \{1, 2\}$, $\{a, c\} = \{1, 2\}$, and $\ell$ is in row $1 + b$. The sides $(2, -2), (1, 1)$ give $\mathsf{A}$, then $(1, -2), (2, 1)$ give $\mathsf{B}$, $(2, -1), (1, 2)$ give $\mathsf{C}$, and $(1, -1), (2, 2)$ give $\mathsf{E}$. □

**Lemma 16** (boundary cross)**.** *Let $h, m \geq 1$. The first row together with the first column of the $h \times m$ grid is a valid blackout. Hence $f(h, m) \geq h + m - 1$.*

*Proof.* If $h = 1$ or $m = 1$ there are no rectangles, so let $h, m \geq 2$. An axis-aligned rectangle on columns $x_1 < x_2$ and rows $y_1 < y_2$ shows $X \times Y$, where $X = \{x_1, x_2\} \setminus \{1\}$ and $Y = \{y_1, y_2\} \setminus \{1\}$ are nonempty. Since $|X| = 1$ forces $x_1 = 1$ and $|Y| = 1$ forces $y_1 = 1$, the set $X \times Y$ determines the rectangle. By Lemma 13 a tilted rectangle has at most two black corners, $\ell$ and the topmost one. If both are black, then $\ell = (1, 1 + b)$ and it shows $p = (1 + c, 1 + b + d)$ and $q = (1 + a + c, 1 + d)$, which determine $a, b, c, d$. Since $p$ and $q$ differ in both coordinates, no axis-aligned rectangle shows $\{p, q\}$. Lemma 4 covers the presentations with three or more points. □

**Lemma 17** (axis-aligned rivals)**.** *Suppose that no axis-aligned rectangle is fully black, as holds when $S$ is axis-valid and $m \geq 3$. Call an axis-aligned rectangle that shows exactly a set $P$ of at most two white points a* rival *of $P$. Then the empty set has no rival, and $P$ has a rival if and only if it is a singleton $\{(x, r)\}$ with $\{r, s\} \in F$ for some $s \in B_x$, a vertical pair $\{(x, r), (x, s)\}$ with $\{r, s\} \in F$, a horizontal pair $\{(x, r), (z, r)\}$ with $B_x \cap B_z \neq \emptyset$, or an oblique pair $\{(x, r), (z, s)\}$ with $x \neq z$, $r \neq s$, and $(x, s), (z, r) \in S$.*

*Proof.* A fully black axis-aligned rectangle is a four-cycle of $G_S$, so the clause after "as holds when" is Lemma 7. A rival of a nonempty $P$ has rows $\{r, s\}$ and columns $\{x, z\}$ as in the statement, with its other corners black. This says $s \in B_x$ and $\{r, s\} \subseteq B_z$ for a singleton, $\{r, s\} \subseteq B_z$ for a vertical pair, $s \in B_x \cap B_z$ for a horizontal pair, and $(x, s), (z, r) \in S$ for an oblique pair. Conversely, as $r \notin B_x$, these conditions force $z \neq x$ and $s \neq r$, and the rectangle on rows $\{r, s\}$ and columns $\{x, z\}$ shows exactly $P$. □

# 5 Three rows

On three rows the row bound of Theorem 8 is attained, and we count the maximum blackouts as words. Throughout, $m \geq 3$, the rows are $1, 2, 3$

from top to bottom, and the tilted rectangles are the diamonds $D(i,1) = \{(i,2),(i+1,1),(i+2,2),(i+1,3)\}$ for $1 \le i \le m-2$ (Lemma 15).

**Theorem 18** (three rows)**.** *For every $m \ge 3$, the maximum blackout of the $3 \times m$ grid has $m+2$ points, that is, $f(3,m) = m+2$.*

*Proof.* The upper bound is the case $h = 3$ of Theorem 8, since $q_3 = 2$. The blackout of row 1 and column 1 has $m+2$ points. It is the boundary cross of Lemma 16 for $h = 3$, so it is valid. □

From now on every blackout has $m+2$ points. We name the columns $a = \{1\}$, $b = \{2\}$, $c = \{3\}$, $A = \{1,2\}$, $B = \{1,3\}$, $C = \{2,3\}$ and $X = \{1,2,3\}$, and call $a$ and $c$ *outer*. A blackout with no empty column is its *word* $B_1B_2\cdots B_m$. We name each pair of $F$ (Section 2) by its double letter.

**Proposition 19** (the two types)**.** *Let $m \ge 3$ and let $S$ be a blackout of the $3 \times m$ grid with $m+2$ points.*

*(a) If $S$ is valid, its word has one letter $X$ and singletons elsewhere (type T), or two different doubles and singletons elsewhere (type P).*

*(b) Conversely, for every word of type T or P the graph $G_S$ is a tree, so no two axis-aligned rectangles have the same presentation.*

*(c) There are exactly $m^2 3^{m-1}$ words of type T or P.*

*Proof.* (a) By Lemma 9, $|S| = t + e(H)$ with $t \le m$ nonempty columns and $e(H) \le 2$ (Lemma 10). So every column is nonempty and $\sum_i (|B_i| - 1) = 2$. Two equal doubles form a four-cycle in $G_S$, which Lemma 7 forbids.

(b) The column of $X$, or the two doubles, connect the three rows, and every other column hangs on one row. So $G_S$ is a tree, and Lemma 7 applies. (c) Types T and P give $m3^{m-1}$ and $\binom{m}{2} \cdot 6 \cdot 3^{m-2} = m(m-1)3^{m-1}$ words. □

We call the words of type T or P *candidates*. They fall into the triple class $X$ and the *ordered classes* $AC$, $CA$, $AB$, $BA$, $BC$ and $CB$, named by the doubles from left to right. The *class* $\{A,C\}$ is $AC \cup CA$, and likewise for $\{A,B\}$ and $\{B,C\}$. Let $AC_m$ be the set of valid words of width $m$ in $AC$, and likewise for the others.

**Lemma 20** (diamond test)**.** *Let $S$ be a candidate with pair set $F$. Then $S$ is valid if and only if every diamond $D(i,1)$ passes the test below, where $P = D(i,1) \setminus S$ is its presentation.*

*(a) If $|P| \geq 3$ or $P$ is empty, the diamond passes.*

*(b) If $P = \{(i+1,1),(i+1,3)\}$, the diamond fails if and only if $B \in F$.*

*(c) If $P = \{(i,2),(i+2,2)\}$, it fails if and only if $B_i \cap B_{i+2}$ meets $\{1,3\}$.*

*(d) If $P$ is an oblique pair $\{(x,r),(z,s)\}$, it fails if and only if $(x,s),(z,r) \in S$.*

*(e) If $P = \{(x,r)\}$, it fails if and only if $\{r,s\} \in F$ for some $s \in B_x$.*

*Proof.* Axis-aligned rectangles are told apart (Proposition 19(b)), and three points fix a rectangle (Lemma 4). Tests (b) to (e) are Lemma 17 for a diamond, since in (b) a rival needs $B \in F$ and in (e) it needs $\{r,s\} \in F$. An empty diamond has no axis-aligned rival, since $G_S$ has no four-cycle. Two diamonds $D(i,1)$ and $D(j,1)$ with $i < j$ share at most one point, so a common presentation has at most one point. Then columns $i+1$ and $j+1$ are both black in rows 1 and 3, which no candidate allows. □

A *window* is a string of three consecutive letters, at columns $i$, $i+1$ and $i+2$. It *passes* or *fails* with the diamond $D(i,1)$. We call $(i,2)$ and $(i+2,2)$ the *tips*. A tip is black exactly when its letter is $b$, $A$, $C$ or $X$.

**Lemma 21** (singleton dichotomy)**.** *Let $S$ be a candidate and let $stu$ be a window of three singletons.*

*(a) If $B \notin F$, the window passes.*

*(b) If $B \in F$, the window fails if and only if it is $bab$, $bbb$ or $bcb$.*

*Proof.* If $s = u = b$, the diamond shows the vertical pair (for $t = b$) or one point of column $i+1$ (for $t$ outer), and tests (b) and (e) fail it exactly when $B \in F$. With one black tip it shows three corners or an oblique pair whose test (d) needs the white point $(i+1,2)$. With no black tip at least three corners show. □

**Lemma 22** (six of nine in the class $\{A,C\}$)**.** *Let $S$ be a candidate of class $\{A,C\}$, so $F = \{A,C\}$, and let $s,t$ be singletons.*

*(a) A window sAt fails if and only if st is bb, bc or cb.*

*(b) A window sCt fails if and only if st is bb, ba or ab.*

*(c) A window with one double, at a tip, passes.*

*Proof.* (a) Here $(i+1,1)$ and $(i+1,2)$ are black and $(i+1,3)$ is white. For $s=t=b$ the diamond shows $(i+1,3)$ alone, and test (e) fails it through $2 \in B_{i+1}$ and $C \in F$. For $s=b \neq t$ it shows $(i+2,2)$ and $(i+1,3)$, and test (d) asks whether $(i+2,3)$ is black, that is, whether $t=c$. The case $t=b \neq s$ is the mirror image, and with no black tip three corners show. Part (b) follows by the reflection in the middle row, which swaps $a, c$ and $A, C$.

(c) Let the window be $Ast$, so the tip $(i,2)$ is black. If $t=b$, the diamond shows the vertical pair or one point of column $i+1$, and both pass by tests (b) and (e), since $B \notin F$. If $t \neq b$, it shows three corners or an oblique pair whose test (d) needs $(i+1,2)$ black, and that point is white. Reflection and reversal give the other windows. □

By Lemma 20, for $m \geq 3$ the set $AC_m$ consists of the words $UAVCW$ of width $m$, with $U$, $V$ and $W$ strings of singletons, whose windows all pass the test with $F=\{A,C\}$. We use the same description for $m=2$, put $G_{AC}(z)=\sum_{m\geq 2}|AC_m|z^m$, and define $G_{AB}$ in the same way from the words $UAVBW$ with $F=\{A,B\}$. The series $G_X(z)=\sum_{m\geq 1}|X_m|z^m$ comes from the words $UXW$ with $F=\{A,B,C\}$, and it starts with the word $X$ at $m=1$. Put $R=1-3z$. The strings of $n \geq 1$ singletons with a prescribed letter next to a double have the generating function $z/R$.

**Proposition 23** (the class $\{A,C\}$)**.** *The generating function of AC is*

$$G_{AC}(z)=\frac{z^2-4z^3+10z^4-8z^5-3z^6}{(1-3z)^3}, \tag{2}$$

*so* $|AC_2|=1$, $|AC_3|=5$ *and*

$$|AC_m|=2(m-1)(3m+2)3^{m-5} \qquad (m \geq 4). \tag{3}$$

*Proof. Step 1. The windows with both doubles.* In a window $tAC$ the points $(i+1,1)$ and $(i+2,2)$ are black and $(i+1,3)$ is white. For $t=b$ the diamond shows $(i+1,3)$ alone, and test (e) fails it. For $t \neq b$ it shows $(i,2)$ and $(i+1,3)$, and test (d) fails it exactly when $t=c$. By the half turn $ACt$ passes only for $t=c$, and $AtC$ passes by tests (b) and (e).

*Step 2. Adjacent doubles.* In $UACW$ every window passes except possibly the two of Step 1 (Lemmas 21(a) and 22(c)). So $U$ ends in $a$ and $W$ begins with $c$, if nonempty, and this case contributes $z^2(1+z/R)^2$.

*Step 3. One singleton between.* In $UAtCW$ the window $AtC$ passes. By Lemma 22(a), the last letter of $U$ has $p_t = 3, 1, 2$ allowed values for $t = a, b, c$, and by (b) the first letter of $W$ has $q_t = 2, 1, 3$. So this case contributes $z^3 \sum_t (1 + p_t z/R)(1 + q_t z/R) = z^3(3 + 12z/R + 13z^2/R^2)$, since $\sum p_t = \sum q_t = 6$ and $\sum p_t q_t = 13$.

*Step 4. At least two singletons between.* In $UAv_1 \cdots v_k CW$ with $k \geq 2$ only $uAv_1$ and $v_k Cw$ can fail. If $U$ is empty, $v_1$ is free, and otherwise six of the nine pairs $(u, v_1)$ pass. So each end contributes $3+6z/R$, the free middle letters give $1/R$, and this case contributes $z^4(3+6z/R)^2/R = 9z^4(1-z)^2/R^3$.

*Step 5. The coefficients.* The three cases sum to (2). The coefficient of $z^m$ in $z^k R^{-3}$ is $\binom{m-k+2}{2} 3^{m-k}$ for $m \geq k$ and 0 otherwise. For $m \geq 4$ the five terms with $k = 2, \ldots, 6$ sum to the polynomial $2(m-1)(3m+2)3^{m-5}$, and at $m = 2, 3$ they give 1 and 5. □

**Theorem 24** (the number of three-row optima)**.** *For $m \geq 3$ let $E_m$ be the number of maximum blackouts of the $3 \times m$ grid whose word is outside $AC$ and $CA$. Then for every $m \geq 4$,*

$$a_3(m) = 4(m-1)(3m+2)3^{m-5} + E_m. \tag{4}$$

*Proof.* Column reversal maps $AC$, $AB$ and $BC$ onto $CA$, $BA$ and $CB$, and the reflection in the middle row maps $AB$ onto $CB$, so

$$a_3(m) = 2|AC_m| + 4|AB_m| + |X_m| = 2|AC_m| + E_m \qquad (m \geq 3), \tag{5}$$

and (3) gives (4). □

A column black in rows 1 and 3 is $B$ or $X$, so $E_m$ is as in Theorem B.

## 5.1 The generating function

Put $L = 1 - 2z - 2z^2$, $N = 1 + 2z^2$ and $D = LN$.

**Theorem 25** (three-row generating function, parts (b) and (c) computer-assisted)**.**

*(a) The series $\sum_{m \geq 3} a_3(m) z^m$ equals $2G_{AC} + 4G_{AB} + G_X - z - 12z^2$. Here $G_{AC}$ is (2), and $G_{AB}$ and $G_X$ are the rational functions of Propositions 53 and 52, with denominators dividing $D^3$ and $D^2$.*

*(b) The reduced denominator of the series is exactly* $Q = R^3L^3N^3$.

*(c) With*

$$\begin{aligned} P(z) = z &- 3z^2 - 34z^3 + 228z^4 - 641z^5 + 1307z^6 - 2186z^7 \\ &+ 2056z^8 - 236z^9 - 1060z^{10} + 2584z^{11} - 4264z^{12} \\ &- 1440z^{13} + 704z^{14} - 1024z^{15} + 2080z^{16} + 2176z^{17} + 384z^{18}, \end{aligned}$$

*we have*

$$\sum_{m\geq 3} a_3(m) z^m = \frac{P(z)}{Q(z)} - z - 12z^2. \tag{6}$$

*Proof.* (a) Use (5). The coefficients 1 and 12 of $z$ and $z^2$ count the word $X$, the six words $sX$ and $Xs$, and the six words of two different doubles.

(b) This is Theorem 54 in Section 10. The orders at the roots of $L$ and $N$ rest on the computer-assisted form (16) of $G_{AB}$.

(c) Clearing denominators in (a) gives $P$, exactly in SymPy (`three_row_closed_form.py`). The script `three_row_closed_form_check.py` checks (6) against an independent transfer matrix with 350 states, in exact integers through width 100. □

**Corollary 26** (asymptotics)**.** *Let* $\rho = 1 + \sqrt{3}$. *As* $m \to \infty$,

$$a_3(m) = \frac{4}{81} m^2 3^m - \frac{4}{243}(m+2)3^m + O(m^2\rho^m). \tag{7}$$

*So the valid fraction* $a_3(m)/(m^2 3^{m-1})$ *of the candidates is* $\frac{4}{27} - \frac{4}{81m} - \frac{8}{81m^2} + O\big((\rho/3)^m\big)$.

*Proof.* The main term of (4) is $4(3m^2-m-2)3^{m-5} = \frac{4}{81}m^2 3^m - \frac{4}{243}(m+2)3^m$. By (5), $E_m$ is the coefficient of $z^m$ in $4G_{AB}+G_X$, whose poles are roots of $D$ of order at most three. The root of least modulus is $1/\rho$, so $E_m = O(m^2\rho^m)$. □

**Theorem 27** (three-row closed form, computer-assisted exact algebra)**.** *Let* $\sigma = 1 - \sqrt{3}$. *For every* $m \geq 3$,

$$E_m = p_+(m)\rho^m + p_-(m)\sigma^m + T_m, \tag{8}$$

*where* $432\,p_\pm(m) = (19 \pm 11\sqrt{3})m^2 + (61 \pm 41\sqrt{3})m + 104 \pm 28\sqrt{3}$ *and* $T_m = (-2)^{\lfloor m/2 \rfloor}\,\tau(m)$ *with* $\tau(m) = (2m^2 + 41m - 104)/216$ *for even* $m$ *and* $\tau(m) = (7m^2 - 32m - 175)/432$ *for odd* $m$. *Hence* $E_m \sim \frac{19+11\sqrt{3}}{432} m^2 \rho^m$, *and* $E_m > 0$.

*Proof.* Let $c_m$ be the coefficient of $z^m$ in $4G_{AB} + G_X$, whose denominator divides $L^3N^3$. By Theorem 25(a), $E_m = c_m$ for $m \geq 3$. The roots $1/\rho$, $1/\sigma$ and $\pm 1/(i\sqrt{2})$ of $L^3N^3$ are triple, so beyond a polynomial part of degree one the $c_m$ have the form (8), and `three_row_closed_form.py` finds the twelve rational coefficients. Both sides satisfy the order-12 recurrence of $L^3N^3$ from $m = 14$ on, so agreement at $m = 2, \ldots, 13$ proves equality for every $m \geq 2$. The asymptotic follows since $|\sigma| < 1$ and $|T_m| = O(m^2 2^{m/2})$. Finally $E_m > 0$, since the cross $Xaa\cdots a$ of Theorem 18 lies in the triple class. □

*Remark* 28 (optima up to symmetry). Let $o_3(m)$ be the number of optima up to the reflection in the middle row and the column reversal. It is 15, 58, 227, 864, 3190 and 11502 at widths 3 to 8. At width 3 the square has four more symmetries, and under all eight there are 10 orbits. By Burnside's lemma $4o_3(m)$ is $a_3(m)$ plus the numbers of optima fixed by the reflection, the reversal and the half turn. An optimum fixed by the reflection uses only $b$, $B$ and $X$, so it has one $X$ and otherwise only $b$. For $m \geq 4$ it contains a window $Xbb$, $bbX$ or $bbb$, which fails since $B \in F$ (Lemmas 20(b) and 21). So the reflection fixes only $bXb$.

Write $m = 2k$ or $m = 2k + 1$. The half turn maps column $i$ to column $m+1-i$ and reflects it, so a word it fixes has its letters $B$ and $X$ in pairs, apart from the central letter of odd $m$. So a fixed optimum with two doubles has $A$ and $C$ at mirror positions, with the central letter $b$ for odd $m$. The reflection maps the fixed words of $AC$ onto those of $CA$, so we count the fixed words in $AC_m$ by the position $j \leq k$ of $A$. Such a word is determined by its first $k$ letters, and $k - 1$ of them are singletons. By Lemmas 21(a) and 22(c) only a window with $A$ or $C$ in the middle can fail, and the half turn maps those at $C$ to those at $A$. If $j = 1$ all $3^{k-1}$ words pass. If $j = k \geq 2$, the window $tAC$ or $tAb$ passes only for $t = a$ (Step 1 of the proof of Proposition 23 and Lemma 22(a)), which leaves $3^{k-2}$ words. For each $1 < j < k$, six of the nine choices of the neighbors of $A$ pass, which leaves $2 \cdot 3^{k-2}$ words. So for $k \geq 2$ the half turn fixes $4k3^{k-2}$ such optima, and for $m = 3$ it fixes $AbC$ and $CbA$. Their series is $W = 2z^3 + \frac{4}{9}(1+z)\big(3z^2(1-3z^2)^{-2} - 3z^2\big)$.

In a palindrome every letter off the center occurs an even number of times. So the reversal fixes no optimum of even width, and for $m = 2k + 1$ only words $UXU^{\mathrm{rev}}$. By Lemma 51 the central window $tXt$ passes only for $t = b$, and the windows $bsX$ and $Xsb$ make the letter at distance two from $X$ outer. Read outward from $X$, the run $U$ is thus counted by $v$ in (12). The half turn fixes the words $UX\iota(U^{\mathrm{rev}})$ of the triple class, where $\iota$ is the reflection. Their

central window has the neighbor pair $(a, c)$, $(b, b)$ or $(c, a)$ and passes, so $U$ is counted by $2u + v$. Adding $z^3$ for $bXb$, $2z\big(u(z^2) + v(z^2)\big)$ and $W$ gives

$$4\sum_{m\geq 3} o_3(m)z^m = \sum_{m\geq 3} a_3(m)z^m + z^3 + \frac{4z^3(1+z^4+z^6)}{M(1+2z^4)} + \frac{2z^3(1-z)(1+5z+3z^2-3z^3)}{(1-3z^2)^2},$$

with $M = 1 - 2z^2 - 2z^4$, since $D(z^2) = M(1 + 2z^4)$. The window tests of `three_row_orbits.py` give the same fixed counts through width 24, and a direct count of the orbits from the definition agrees for $m \leq 10$.

# 6 Four rows

On four rows the row bound of Theorem 8 is $m+4$, attained for every $m \geq 6$. We write $(m)_k = m(m-1)\cdots(m-k+1)$ and use the tilted rectangles $D(i, 1)$, $D(i, 2)$, $\mathsf{A}_i$, $\mathsf{B}_i$, $\mathsf{C}_i$, $\mathsf{E}_i$ of Lemma 15.

**Theorem 29** (four rows)**.** *For every $m \geq 6$ we have $f(4, m) = m + 4$.*

The upper bound is Theorem 8 with $h = 4$. For the lower bound we take

$$S = \{(i, 1) : 2 \leq i \leq m-1\} \cup \{(m-1, 2), (m, 2), (1, 3), (2, 3), (1, 4), (m, 4)\}, \tag{9}$$

which has $m+4$ points. The double columns are column 1 (rows $3, 4$), column 2 (rows $1, 3$), column $m-1$ (rows $1, 2$), and column $m$ (rows $2, 4$). So $G_S$ is the eight-cycle $r_1\, c_2\, r_3\, c_1\, r_4\, c_m\, r_2\, c_{m-1}$ with the columns $c_3, \ldots, c_{m-2}$ hanging off $r_1$.

*Proof of Theorem 29. Step 1. Axis-aligned rectangles.* The graph $G_S$ has no cycle of length four or six. So by Lemma 7 the axis-aligned rectangles have distinct presentations, and none is hidden.

*Step 2. The weak tilted rectangles.* By Lemma 15, a corner of a tilted rectangle in row 1 is always black, and one in row 4 is always white. The other black corners are $(i + 2, 2)$ for $i \geq m - 3$ and $(i + 1, 3)$ for $i = 1$ in $D(i, 1)$, and $(i, 3)$ for $i \leq 2$ and $(i + 1, 2)$ for $i = m - 2$ in $D(i, 2)$. In $\mathsf{A}_i$ and $\mathsf{B}_i$ they are $(i, 3)$ for $i \leq 2$ and $(i + 3, 2)$ for $i \geq m - 4$, and in $\mathsf{C}_i$ and $\mathsf{E}_i$ there are none. As $m \geq 6$, the two conditions for $D(i, 2)$ exclude each other. A rectangle with at most one black corner is safe by Lemma 4. So only the rectangles below need a check, with $s = 1$ for $\mathsf{A}_i$ and $s = 2$ for $\mathsf{B}_i$.

| rectangle | which $i$ | corners shown |
|---|---|---|
| $D(1,1)$ | | $(1,2),(3,2)$ |
| $D(i,1)$ | $i \in \{m-3, m-2\}$ | $(i,2),(i+1,3)$ |
| $\mathsf{A}_i$ or $\mathsf{B}_i$ | $i \in \{1,2\}$, $i \notin \{m-4, m-3\}$ | $(i+s,4),(i+3,2)$ |
| $\mathsf{A}_i$ or $\mathsf{B}_i$ | $i \in \{m-4, m-3\}$, $i \notin \{1,2\}$ | $(i,3),(i+s,4)$ |
| $\mathsf{A}_i$ or $\mathsf{B}_i$ | $i = 2$, $m = 6$ | $(i+s,4)$ |

*Step 3. Against each other.* The rows of the shown points are $2,2$ and $2,3$ and $4,2$ and $3,4$ in the first four lines, so different lines never agree. Within a line the point in row 2 or 3 determines $i$, and the point in row 4 then determines $s$.

*Step 4. Against axis-aligned rectangles.* A rival of $\{(1,2),(3,2)\}$ needs a row $r$ with $(1,r)$ and $(3,r)$ black, but there is none. In the next three lines the axis-aligned rectangle through the shown points has the white corner $(i,3)$, $(i+s,2)$, or $(i,4)$, respectively, since $i \geq 3$, $i+s \leq 4 < m-1$, and $3 \leq i \leq m-3$ there. In the last line $m = 6$, and the shown point is $(z,4)$ with $z = i+s \in \{3,4\}$. Column $z$ is black only in row 1, so a rival needs a column black in rows 1 and 4, and there is none. By Steps 1 and 2 no rectangle is hidden, so $S$ is valid. □

**Lemma 30** (equality in the row bound on four rows)**.** *Let $m \geq 4$. A blackout of the $4 \times m$ grid with $m+4$ points is axis-valid if and only if it has the form described next. Exactly four columns have two black points, their row pairs are the edges of a four-cycle on the rows, and every other column has one black point. There are $3\,(m)_4\,4^{m-4}$ such blackouts, which is 72 and 1,440 for $m = 4$ and 5.*

*Proof.* This is Theorem 11 with $h = 4$. There are three four-cycles on the rows, $(m)_4$ ways to place their labeled edges, and four rows for each other column. □

**Corollary 31** (widths four and five, computer-assisted)**.** *We have $f(4,4) = 7$ and $f(4,5) = 8$.*

*Proof.* The lower bounds are Lemma 16. For the upper bounds, the script `verify_narrow.py` finds a repeated presentation in each of the 72 and 1,440 candidates of Lemma 30. The brute-force check of Section 9 confirms both values without the lemma. □

**Theorem 32** (no hidden rectangle)**.** *Let $m \geq 4$. A valid blackout of the $4 \times m$ grid hides no rectangle.*

*Proof.* By Lemma 2, two rectangles have the same presentation exactly when the symmetric difference of their corner sets is black. So it suffices to write each corner set as $C(R_1) \triangle C(R_2)$ with $R_1 \neq R_2$. For the axis-aligned rectangle on rows $r, s$ and columns $x, y$, take rows $r, s$ with columns $\{x, w\}$ and $\{y, w\}$ for a third column $w$. For $1 \leq i \leq m-3$ let $\mathsf{Q}_i = \{(i,2), (i,3), (i+3,2), (i+3,3)\}$. Checking corners gives

$$\begin{aligned} \mathsf{C}_i &= \mathsf{A}_i \triangle \mathsf{Q}_i, & \mathsf{E}_i &= \mathsf{B}_i \triangle \mathsf{Q}_i, \\ D(i+1,1) &= D(i,2) \triangle \mathsf{A}_i, & D(i+1,2) &= D(i,1) \triangle \mathsf{E}_i, \end{aligned} \tag{10}$$

and each identity can be solved for each of its three rectangles. The first line handles $\mathsf{A}_i$, $\mathsf{B}_i$, $\mathsf{C}_i$, and $\mathsf{E}_i$. A diamond $D(j,1)$ or $D(j,2)$ has $j \geq 2$ or $j \leq m-3$, since $m \geq 4$, and the second line with $i = j-1$ or $i = j$ handles it. □

The hypothesis $m \geq 4$ is needed, since the valid blackout $\{(1,2), (2,1), (3,2), (2,3)\}$ of the $4 \times 3$ grid hides $D(1,1)$, as the Lean kernel checks (Section 9).

Let $m \geq 6$. By Lemma 30 an optimum has four double columns, whose pairs form a four-cycle $F$ on the rows, and singletons elsewhere. We name $F$ by its parts, $\{1,2\} \mid \{3,4\}$, $\{1,3\} \mid \{2,4\}$, or $\{1,4\} \mid \{2,3\}$. A *block* is a run of four consecutive singleton columns, and its *word* lists their black rows. The *template words* 2413, 3412, 2143, and 3142 are the corner rows of $\mathsf{C}_x$, $\mathsf{A}_x$, $\mathsf{E}_x$, and $\mathsf{B}_x$. A *swap word* exchanges two letters of a template word, for example 4213. An *edit word for $F$* replaces one letter $r$ of a template word by a row $s$ with $\{r, s\} \in F$.

**Lemma 33** (forbidden blocks)**.** *Let $S$ be a valid blackout of the $4 \times m$ grid with $m+4$ points and pair set $F$, where $m \geq 6$. No block of $S$ has a swap word or an edit word for $F$. There are* 20 *swap words, and* 24, 24, *or* 32 *edit words for the three pair sets, none of them a swap word. So at most* 212 *of the* 256 *words of a block are allowed.*

*Proof.* Let the block start in column $x$, and let $T$ be one of $\mathsf{C}_x$, $\mathsf{A}_x$, $\mathsf{E}_x$, $\mathsf{B}_x$, with template word $t_1 t_2 t_3 t_4$. In both steps an axis-aligned rectangle has the same presentation as $T$. *Step 1. Swap words.* If the letters in positions $p \neq q$ are exchanged, $T$ shows exactly $(x+p-1, t_p)$ and $(x+q-1, t_q)$, and so does the axis-aligned rectangle on these two points. The template words are the permutations with both end letters in $\{2,3\}$. Exchanging two end letters, or

two middle letters, gives a template word again, and exchanging an end letter with a middle letter gives every permutation with exactly one end letter in $\{2,3\}$. So the 20 swap words are the four template words and the sixteen permutations with exactly one end letter in $\{2,3\}$.

*Step 2. Edit words.* If $t_p = r$ is replaced by $s$ with $\{r,s\} \in F$, then $T$ shows only $(x+p-1, r)$. So does the rectangle on rows $r, s$ and the columns $x+p-1$ and $d$, where $d$ is the double column black in rows $r, s$. Each template word has 8 edit words.

Two template words share an edit word only if they differ in two positions, carrying letters $r, s$ in opposite orders, and then they share two if $\{r,s\} \in F$. This happens for $2413, 3412$ and $2143, 3142$ with $\{r,s\} = \{2,3\}$, and for $2413, 2143$ and $3412, 3142$ with $\{r,s\} = \{1,4\}$. So there are $32 - 4[\{2,3\} \in F] - 4[\{1,4\} \in F]$ edit words. They repeat a letter, so they are not swap words. □

**Theorem 34** (forbidden-block bound)**.** *For every $m \geq 6$,*

$$a_4(m) \leq 3\,(m)_4\,4^{m-4}\,(53/64)^k = O\big(m^4\,212^{m/4}\big), \tag{11}$$

*where $k = \max(0, \lfloor m/4 \rfloor - 4)$.*

*Proof.* Fix $F$, the double columns, and their pairs, in one of $3\,(m)_4$ ways. Of the $\lfloor m/4 \rfloor$ groups of columns $4j-3, \ldots, 4j$, at most four contain a double column. So at least $k$ are blocks, each with at most $212 = 4^4 \cdot 53/64$ words by Lemma 33. Every other column has at most 4 values, which gives (11). □

So the valid fraction of the candidates tends to zero (compare Corollary 26).

**Lemma 35** (six columns of memory)**.** *Fix a pair set $F$ and let $m \geq 6$. A word over the four singleton letters and the pairs of $F$, with each pair used once, is the word of an optimum if and only if no tilted rectangle is hidden, no tilted rectangle shares its presentation with an axis-aligned rectangle whose columns are among its own columns and the double columns, and no two tilted rectangles inside seven consecutive columns share a presentation.*

*Proof.* An optimum satisfies the three conditions, by Theorem 32 for the first. Conversely, by Lemma 30 the word is axis-valid, and it hides no axis-aligned rectangle, since that would be a four-cycle of $G_S$. With the first condition no rectangle is hidden. So two distinct rectangles with the same presentation $P$

have $P \neq \emptyset$, and at least one of them is tilted. If the other is axis-aligned, each of its columns either meets $P$, and so is a column of the tilted one, or has two black corners and is a double column. The second condition excludes this. If both are tilted, they share a column $c$ that meets $P$. Each meets at most four consecutive columns, so both lie in the columns $c-3$ to $c+3$, and the third condition excludes this. □

By Lemma 17 the second condition is decided by the columns of the tilted rectangle and $F$. So a finite automaton that reads a blackout column by column and remembers its last six columns and the pairs of $F$ already used accepts exactly the optima with pair set $F$, and their generating function is rational (Flajolet and Sedgewick [14, Section V.5]).

**Proposition 36** (small widths, computer-assisted)**.** *In the row $h = 4$ of* A399580*, $a_4(4) = 316$, $a_4(5) = 1{,}810$, $a_4(6) = 26$, and $a_4(7) = 292$.*

*Proof.* The ILP program and the brute-force check of Section 9 both find these values. For $m = 6, 7$ the script `four_row_direct.py` tests the candidates of Lemma 30, and the transfer counter `four_row_counts.py` also gives 26 and 292 (Table 1). Its automaton is the exported machine read by `four_row_machines.py`. The script `four_row_equivalence_check.py` builds the automaton of Lemma 35 from the definition and checks that the two automata accept the same words. □

| $m$ | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|
| $a_4(m)$ | 26 | 292 | 2,118 | 13,332 | 75,398 | 402,656 | 2,037,848 |
| $\{1,2\} \mid \{3,4\}$ | 0 | 4 | 128 | 920 | 4,852 | 24,112 | 114,424 |
| $\{1,3\} \mid \{2,4\}$ | 20 | 256 | 1,780 | 11,128 | 63,540 | 346,320 | 1,785,768 |
| $\{1,4\} \mid \{2,3\}$ | 6 | 32 | 210 | 1,284 | 7,006 | 32,224 | 137,656 |

Table 1: Four-row optimum counts by pair set, computer-assisted. Two transfer counters on the same automaton agree to width 1500 (data file `four_row_counts.json.gz`), and a direct test of every candidate agrees to width 10.

**Theorem 37** (the four-row generating function, computer-assisted)**.** *There are coprime integer polynomials $P_4 = 26z^6 - 826z^7 + \cdots$ of degree 444 and $Q_4 = \sum_{j=0}^{420} q_j z^j$ of degree 420 with $q_0 = 1$ and $\sum_{m \geq 6} a_4(m) z^m = P_4/Q_4$. The*

*recurrence* $\sum_j q_j\, a_4(n-j) = 0$ *holds for every* $n \geq 445$ *and fails at* $n = 444$. *Over the integers*

$$Q_4 = (1-z)^2(1+z)^2\, g_5^3\, \tilde{g}_5^3\, g_6^3\, g_{11}^5\, g_{17}^3\, g_{19}^3\, g_{20}^5\, g_{21}^5,$$

*where* $g_d$ *and* $\tilde{g}_d$ *are irreducible of degree* $d$ *with constant term* 1, *and all of* $g_{20}$ *comes from the pair set* $\{1,3\} \mid \{2,4\}$.

*Proof.* The Berlekamp–Massey algorithm modulo eight primes, run on the values $a_4(m)$ for $m \leq 1500$ from the automaton of Proposition 36, gives $P_4$ and $Q_4$. With a register of used pairs the automaton has a transfer matrix $T_F$ with 6,128, 7,792, or 4,912 product states, of which 2,519, 3,089, and 2,119 are reachable. Its state vectors $v(n) = v(0)T_F^n$ satisfy $\sum_j q_j v(n_F - j) = 0$ at $n_F = 436$, 435, and 445. Multiplying by powers of $T_F$ gives the recurrence for every $n \geq 445$, and $\deg P_4 = 444$ forces the failure at $n = 444$. We factor $Q_4$ in SymPy, and `four_row_recurrence.json` lists the factors. □

**Corollary 38** (growth, computer-assisted)**.** *Let* $\rho_4 = 3.351996777310\ldots$ *be the largest real root of the irreducible polynomial* $p(x) = x^{20} g_{20}(1/x) = x^{20} - 4x^{19} + 2x^{18} + \cdots - 26x^2 - 4x + 8$, *and let* $r = 1/\rho_4$. *Put* $H(z) = P_4(z)(1 - z/r)^5/Q_4(z)$, $c_j = (-r)^{5-j} H^{(5-j)}(r)/(5-j)!$, *and* $B(m) = \sum_{j=1}^{5} c_j \binom{m+j-1}{j-1}$. *Then* $\deg B = 4$ *and* $a_4(m) = \rho_4^m B(m) + O((1000/341)^m)$, *with a large constant. So* $a_4(m) \sim c\, m^4 \rho_4^m$ *with* $c = c_5/24 \approx 8.1611 \times 10^{-5}$.

*Proof.* The script `four_row_root_isolation.py` isolates the complex roots of every factor of $Q_4$ in rational rectangles, in exact arithmetic in SymPy. The only root of modulus at most 0.341 is $r = 0.29832\ldots$, which is simple in $g_{20}$ and so has order five in $Q_4$. It is not a root of $P_4$, since the two are coprime. So $H$ is analytic at $r$ and $c_5 = H(r) \neq 0$. Taylor expansion of $H$ at $r$, with $z - r = -r(1 - z/r)$, shows that the principal part of $P_4/Q_4$ at $r$ is $\sum_{j=1}^{5} c_j (1 - z/r)^{-j}$. What remains is a polynomial plus a rational function with no pole in $|z| \leq 0.341$, so its coefficients are $O((1000/341)^m)$ by the Cauchy bound. As $[z^m](1 - z/r)^{-j} = \rho_4^m \binom{m+j-1}{j-1}$, this gives the expansion, and $B$ has leading coefficient $c_5/24$. Interval arithmetic on a rational interval of width below $2^{-700}$ around $r$ puts $c$ between 0.000081610970775 and 0.000081610970776. □

# 7 Five and six rows

One finite padding test settles every large width on five and on six rows.

## 7.1 A finite test for boundary padding

For a blackout $S$ of the $h \times m$ grid and $L, R \geq 0$, let $S_{L,R}$ add $L$ columns on the left and $R$ on the right. They are the columns $1-L, \ldots, 0$ and $m+1, \ldots, m+R$. Each is black exactly in the *padding row*, which is row 1 or row $h$. As before, $W_h$ is the span bound of Lemma 14.

**Theorem 39** (finite collar)**.** *Let $h \geq 3$ and $m \geq 1$, and let $S$ be a blackout of the $h \times m$ grid. Fix the padding row and choose the left side, the right side, or both. Call $(L, R)$* admissible *if $L = 0$ when the left side is not chosen and $R = 0$ when the right side is not chosen. Put $L^* = W_h$ if the left side is chosen and $L^* = 0$ otherwise, and define $R^*$ in the same way. Then $S_{L,R}$ is valid for every admissible $(L, R)$ if and only if $S_{L^*,R^*}$ is valid.*

*Proof.* Write $S^* = S_{L^*,R^*}$. If every admissible padding is valid, then so is $S^*$. Conversely, let $S^*$ be valid and $U = S_{L,R}$ with $(L, R)$ admissible. By Lemma 13 the corners of a tilted rectangle lie in the rows $j$, $j-b$, $j+d$, $j+d-b$ with $b, d \geq 1$, so at most one lies in row 1 and at most one in row $h$. Hence a weak tilted rectangle of $U$ has a black corner in an original column $e$, and by Lemma 14 its columns lie in $[e-W_h, e+W_h] \cap [1-L, m+R] \subseteq [1-L^*, m+R^*]$. Since $U$ and $S^*$ agree on their common columns, it is a rectangle of $S^*$ with the same presentation. Suppose that two distinct rectangles $R_1$ and $R_2$ of $U$ have the same presentation $P$.

*Step 1. Both axis-aligned.* The graph $G_U$ is $G_S$ plus one vertex of degree one for each added column, so its cycles are cycles of $G_{S^*}$. Since $S^*$ is valid with at least three columns, neither graph has a cycle of length four or six, and $U$ is axis-valid (Lemma 7).

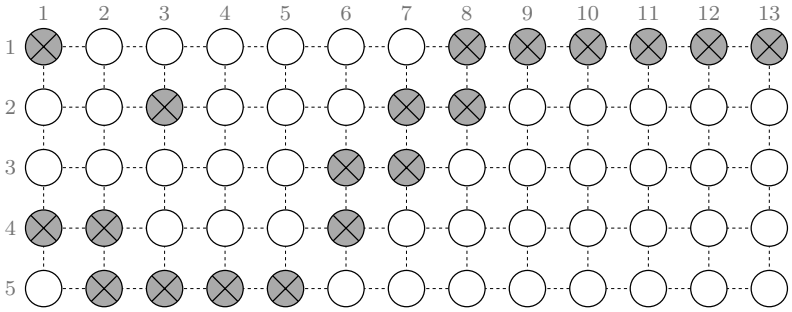
*Step 2. Both tilted.* By Lemma 4, $|P| \leq 2$, so both are weak and are rectangles of $S^*$ with the same presentation, which is impossible.

*Step 3. One of each.* Let $R_1$ be tilted, so it is a rectangle of $S^*$. A column of $R_2$ with a point of $P$ is a column of $R_1$, and one without has two black corners and is an original column. So $R_2$ is a rectangle of $S^*$ too, a contradiction. □

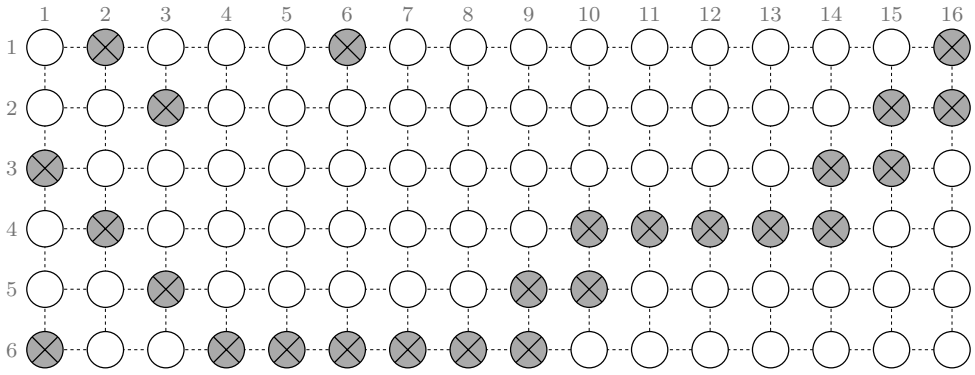
*Remark* 40 (the padding row has to be a boundary row). Take $h = 3$, $B_1 = B_2 = \{1\}$, $B_3 = \{1, 3\}$, and the interior padding row 2. The added columns 0 and $-1$ keep the board valid, but $-2$ breaks it. The diamond $(-2, 2)$, $(-1, 1)$, $(0, 2)$, $(-1, 3)$ and the rectangle on columns $-1$, 3 and rows 1, 3 both show $\{(-1, 1), (-1, 3)\}$.

## 7.2 Five rows

Let $S_8$ be the blackout of the $5 \times 8$ grid with $B_1 = \{1, 4\}$, $B_2 = \{4, 5\}$, $B_3 = \{2, 5\}$, $B_4 = B_5 = \{5\}$, $B_6 = \{3, 4\}$, $B_7 = \{2, 3\}$, and $B_8 = \{1, 2\}$. Its doubles carry the edges of $K_{2,3}$ between $\{2, 4\}$ and $\{1, 3, 5\}$. With five more columns black in row 1 it gives the $5 \times 13$ board of Figure 2(a).




(a)




(b)

Figure 2: The $5 \times 13$ test board of $S_8$ in (a) and the blackout $S_{16}$ of the $6 \times 16$ grid in (b). Gray crossed circles are blacked out and white circles are visible, and rows are numbered from the top.

**Theorem 41** (five rows, computer-assisted)**.** *(a) We have* $f(5, m) = m + 5$ *for* $m = 5$*,* $6$*, and* $7$*.*

*(b) We have* $f(5, m) = m + 6$ *for every* $m \geq 8$*, attained by* $S_8$ *followed by* $m - 8$ *columns black in row* $1$ *only.*

*(c) The $5 \times 8$ grid has exactly four maximum blackouts, the images of $S_8$ under the row reflection and the column reversal.*

*Proof.* (b) The upper bound is Theorem 8. Apply Theorem 39 to $S_8$ with the padding row 1 on the right. Since $W_5 = 5$, the test board is the $5 \times 13$ board of Figure 2(a). Three rectangle enumerators list its 954 rectangles and find no repeated presentation.

(a) The lower bounds are $S_0$ of Theorem 46 below for $m = 5$ and the stored blackouts of $h + m$ points (Section 9) for $m = 6, 7$. For $m = 5$ the upper bound is Corollary 12. For $m = 6$ and $m = 7$ a valid blackout of $m + 6$ points would be one of the $10\,(m)_6\,5^{m-6}$ candidates of Theorem 11. Two SAT encodings exclude all of them. Two more encodings and a column search, which use no structure theorem, agree.

(c) Among the $10\,(8)_6\,5^2 = 5{,}040{,}000$ candidates of Theorem 11, two SAT enumerations and a column search find exactly the four images of $S_8$. □

*Remark* 42 (five-row counts, computer-assisted). Table 2 gives $a_5(m)$ for $8 \le m \le 12$. For $m = 13, 14, \ldots, 23$ the numbers $a_5(m)$ are 193,406, 1,346,758, 8,650,416, 52,128,242, 299,349,210, 1,653,637,492, 8,828,301,226, 45,743,165,894, 230,934,909,328, 1,139,486,126,866, and 5,510,071,794,736. The values for $m \le 12$ come from two independent programs and the others from two exact transfer counters, all based on Theorem 11. No proof uses them.

## 7.3 Six rows and small grids

Let $S_{16}$ be the blackout of the $6 \times 16$ grid in Figure 2(b). Its doubles, in the columns $1, 2, 3, 6, 9, 10, 14, 15, 16$, carry the edges of the complete bipartite graph between $\{1, 3, 5\}$ and $\{2, 4, 6\}$.

**Theorem 43** (six rows, computer-assisted)**.** *(a) For $8 \le m \le 14$ we have $f(6, m) \le m + 8$.*

*(b) For every $m \ge 15$ we have $f(6, m) = m + 9$.*
*So for $m \ge 8$ we have $f(6, m) = m + 9$ if and only if $m \ge 15$.*

*Proof.* (a) For $m = 8$ this is Corollary 12. For $9 \le m \le 14$, by Theorem 11 the doubles of a valid blackout of $m + 9$ points carry the edges of $K_{3,3}$ for one of ten row partitions. Two SAT encoders find none for any width and

partition. A column search for $m \le 13$ and a third encoder for $m = 13$ and $m = 14$ give a third check at every width.

(b) The upper bound is Theorem 8. A valid blackout of 24 points on the $6 \times 15$ grid is in `jis/data/six_row_sat_w15_A.json`. For $m \ge 16$ apply Theorem 39 to $S_{16}$ with the padding row 1 on the right. Since $W_6 = 7$, the test board adds $(17, 1), \ldots, (23, 1)$, and the three enumerators find no repeated presentation among its 4,627 rectangles. □

**Theorem 44** (the guess fails, computer-assisted)**.** *(a) We have* $f(6, 6) = 12$*,* $f(6, 7) = 13$*,* $f(7, 7) = 14$*, and* $f(6, 8) = 15$*. The blackout* $(1, 1)$*,* $(1, 5)$*,* $(2, 1)$*,* $(3, 1)$*,* $(3, 2)$*,* $(3, 3)$*,* $(4, 3)$*,* $(4, 4)$*,* $(5, 4)$*,* $(6, 4)$*,* $(7, 4)$*,* $(7, 5)$*,* $(7, 6)$*,* $(8, 2)$*,* $(8, 6)$ *of the* $6 \times 8$ *grid attains the last value.*

*(b) The guess of Section 1 fails. Indeed* $f(5, 8) = 14$ *and* $f(6, 8) = 15$*.*

*Proof.* (a) The other lower bounds are stored blackouts of $h + m$ points. Encoding A of Section 9 on CaDiCaL and encoding B on Glucose and MiniSat find no valid blackout of 13, 14, 15, or 16 points on the $6 \times 6$, $6 \times 7$, $7 \times 7$, or $6 \times 8$ grid, respectively. A third encoding agrees on the last three, and the ILP program on the first. On the $7 \times 7$ and $6 \times 8$ grids the third encoding splits the search into cases by the first row with at least three black points.

(b) The values are Theorem 41(b) and part (a). The guess gives 13 and 14 on these grids. The blackout $S_8$ has 14 points and the blackout of part (a) has 15, and a check of their 364 and 592 rectangles shows that both are valid. So the guess fails, and this uses no solver. □

If a side is at most four, then $f(h, m) \le h + m$ by Sections 3 to 6. Also $f(h, m) = h + m$ on $5 \times 5$, $5 \times 6$, $5 \times 7$, and $6 \times 6$, the other grids of area below 40. So $f$ first exceeds $h + m$ on the $5 \times 8$ grid.

Neuwirth [32, Remark 18] states the case $5+5$ of the next lemma without proof.

**Lemma 45** (few edges without short cycles)**.** *Let* $G$ *be a finite simple bipartite graph with* $v$ *vertices,* $1 \le v \le 10$*, and with no cycle of length four or six. Then* $G$ *has at most* $v$ *edges. If it has exactly* $v$ *edges, then* $G$ *is connected and has exactly one cycle, of length* 8 *or* 10*.*

*Proof.* Every cycle of $G$ has length at least 8. Suppose a component has more edges than vertices. Removing vertices of degree one with their edges until none is left gives a connected $H$ with more edges than vertices and minimum

degree two, so some vertex $x$ of $H$ has degree at least three. Let $D_k$ be the set of vertices at distance $k$ from $x$. Two paths of length at most 3 from $x$ to one vertex would close a cycle of length at most 6. So for $k = 1, 2, 3$ every vertex of $D_k$ has exactly one neighbor in $D_{k-1}$ and none in $D_k$. Its degree is at least two, so it also has a neighbor in $D_{k+1}$. For $k \leq 2$ no other vertex of $D_k$ shares that neighbor. Hence $|D_1| \geq 3$, $|D_3| \geq |D_2| \geq |D_1|$ and $|D_4| \geq 1$, at least 11 vertices in all. So $G$ has at most $v$ edges, and with exactly $v$ edges every component has one cycle. Two disjoint cycles of length at least 8 need 16 vertices, so $G$ is connected. □

**Theorem 46** (the $5 \times 5$ grid)**.** *(a) We have $f(5,5) = 10$, and $G_S$ is connected with exactly one cycle for every optimum $S$.*

*(b) (Computer-assisted.) There are exactly $a_5(5) = 8$ maximum blackouts, the eight distinct images of $S_0$ below under the symmetries of the square.*

*Proof. Upper bound.* If $S$ is valid, $G_S$ has 10 vertices, $|S|$ edges and no cycle of length four or six (Lemma 7), so Lemma 45 applies.

*Lower bound.* Let $S_0$ be the staircase of nine points plus the corner $(5,5)$ with $B_1 = \{4,5\}$, $B_2 = \{3,4\}$, $B_3 = \{1,2,3\}$, $B_4 = \{1\}$, and $B_5 = \{1,5\}$. Its graph is an 8-cycle with two pendant edges, so $S_0$ is axis-valid. By Lemma 13 the 30 tilted rectangles among the 130 are the nine unit diamonds, the four kinds of Lemma 15 in four positions each, and five that span the whole grid, one of them a large diamond. Exactly fifteen are weak. The table lists the two corners each one shows and, after a bar for an oblique pair, the white corner of its axis-aligned rectangle.

| | | | |
|---|---|---|---|
| $(1,2),(2,1) \mid (1,1)$ | $(1,3),(2,2) \mid (1,2)$ | $(3,4),(4,3) \mid (4,4)$ | $(2,5),(3,4) \mid (3,5)$ |
| $(2,2),(4,2)$ | $(1,2),(4,3) \mid (1,3)$ | $(4,4),(5,2) \mid (4,2)$ | $(3,5),(4,4) \mid (3,4)$ |
| $(4,3),(5,2) \mid (4,2)$ | $(3,4),(5,2) \mid (5,4)$ | $(2,5),(5,2) \mid (2,2)$ | $(4,5),(5,3) \mid (4,3)$ |
| $(1,3),(4,2) \mid (1,2)$ | $(2,5),(4,3) \mid (4,5)$ | | $(4,5),(5,4) \mid (4,4)$ |

The fifteen pairs are distinct, and the other tilted rectangles are determined by Lemma 4. Each listed white corner lies on the only axis-aligned rectangle through its pair, and $(2,2),(4,2)$ has no rival since $B_2 \cap B_4 = \emptyset$. So $S_0$ is valid.

*(b).* Of the 44,640 sets of 10 points without a short cycle, the script `verify_narrow.py` finds exactly the 8 images of $S_0$ valid. The ILP program and the brute-force check agree. □

# 8 Every fixed height

Fix $h$. A blackout of the $h \times m$ grid is a *column word* $w = B_1 B_2 \cdots B_m$ over the subsets of $\{1, \ldots, h\}$, with length $|w|$ and blackout $S(w)$. Let $\mathcal{L}_h$ be the set of words with a valid blackout, the empty word included. A *factor* of $w$ is a word $B_i B_{i+1} \cdots B_j$.

**Proposition 47** (validity is not a window property)**.** *Let $g \geq 1$. The blackout $S = \{(1,2), (3,2), (g+3,1), (g+3,3)\}$ of the $3 \times (g+3)$ grid is not valid, but every factor of at most $g$ columns is valid.*

*Proof.* The diamond $D(1,1)$ and the rectangle on columns $2, g+3$ and rows $1, 3$ both show $\{(2,1), (2,3)\}$, while a factor of at most $g$ columns has at most two black points and two different rectangles differ in at least four corners (Lemmas 2 and 4). □

**Theorem 48** (regularity and rational series)**.** *For every $h \geq 1$ the language $\mathcal{L}_h$ is regular, and $G_h(z,u) = \sum_{w \in \mathcal{L}_h} z^{|w|} u^{|S(w)|}$ is rational in $z$ and $u$.*

*Proof.* The *trace* of a rectangle $R$ is the word whose $i$th letter is the set of rows of its corners in column $i$, and it determines $C(R)$. By Lemma 2, $w \notin \mathcal{L}_h$ exactly when two different traces $Y, Y'$ of length $|w|$ have $Y_i \triangle Y'_i \subseteq B_i$ for every $i$. The traces form a regular language, $\emptyset^*\{r,s\}\emptyset^*\{r,s\}\emptyset^*$ for axis-aligned rectangles and, by Lemmas 13 and 14, $\emptyset^* Y_0 Y_1 \cdots Y_n \emptyset^*$ for finitely many templates (1) of span $n \leq W_h$. So a finite automaton that reads $w$ and two traces in parallel recognizes the complement of $\mathcal{L}_h$, and $\mathcal{L}_h$ is regular. With the weight $u^{|B|}$ on a letter $B$, the transfer matrix $M(u)$ of a deterministic automaton for $\mathcal{L}_h$ gives $G_h = \alpha(I - zM(u))^{-1}\beta$, rational by Cramer's rule [14, Proposition V.6]. □

**Theorem 49** (fixed height)**.** *Let $h \geq 2$. (a) There are integers $m_0 \geq 3$, $p \geq 1$, and $k_0, \ldots, k_{p-1}$ such that $f(h,m) = m + k_{m \bmod p}$ for every $m \geq m_0$. (b) The series $\sum_{m \geq 0} a_h(m) z^m$ is rational, with the conventions of Section 2, and the words of optimum blackouts form a regular language.*

*Proof.* Let $E(w) = |S(w)| - |w|$. Theorem 8 and $|S(v)| \leq h|v|$ bound the excess of every factor of a valid word by $C_h = \max(q_h, 2h-2)$. So a counter for the excess of prefixes, kept in $[k - C_h, C_h]$, gives an automaton for the valid words $\mathcal{L}_{h,k}$ of excess $k$, and their lengths are eventually periodic. For $m \geq 3$, Lemma 16 and Theorem 8 make $f(h,m) - m$ the largest $k$ from $h-1$

to $q_h$ with a word of length $m$ in $\mathcal{L}_{h,k}$, which gives (a). The optimum words are those of $\mathcal{L}_{h,k_r}$ of lengths $m \geq m_0$ with $m \equiv r \pmod{p}$, and finitely many shorter ones, so (b) follows as in Theorem 48. □

**Corollary 50** (fixed deficit)**.** *Let $h \geq 2$, and let $d \geq 0$ be an integer. The words of valid blackouts with exactly $f(h,m) - d$ points form a regular language with a rational counting series, and so do those with at least $f(h,m) - d$ points.*

*Proof.* If $w = uv$ is valid of excess $k$, then $k - C_h \leq E(u) = k - E(v) \leq C_h$ for every integer $k$, so the counter of Theorem 49 makes $\mathcal{L}_{h,k}$ regular. For $m \geq m_0$ with $m \equiv r \pmod{p}$, the two languages at length $m$ are $\mathcal{L}_{h,k_r-d}$ and the union of $\mathcal{L}_{h,k}$ over $k_r - d \leq k \leq k_r$. We take these at those lengths, unite over $r$, and add finitely many shorter words. □

The corollary is for fixed $h$ and $d$ and says nothing about counts up to symmetry. Guilhot, Little, and Parkinson [18, Theorem 1(3)] prove that the words of maximal weight in a regular language form a regular language when the additive weight is bounded above. With our bounds this also gives (a), (b), and Corollary 50. On valid words of length at least 3 the excess $E$ is an integer at most $q_h$, and an optimum has $E \geq h - 1$. So finitely many removals of the words of maximal excess give each $\mathcal{L}_{h,k}$ with $k \geq h - 1 - d$, and a word of level $k$ has deficit $d$ exactly when $k + d$ is the highest level that has its length ($d = 0$ gives the optima). Our proof builds the automaton from the rectangles.

# 9 Computations and verification

The results that rest on a computation are labeled computer-assisted. Every other result in Sections 2 to 8 has a written proof. Every computed value and count comes from at least two programs written separately, which agree wherever both ran, with the one exception of the $7 \times 8$ count below.

All programs are in Python 3.14.3. The ILP program `solver/engine.py` finds a smallest hitting set for the difference sets of Lemma 2 with CBC 2.10.3 [15] through PuLP 3.3.2 [30]. The brute-force check, `solver/brute.py`, uses no lemma and shares no code with it. The verifier `verify_narrow.py` checks the lemmas and constructions of Sections 4 to 6 and the $5 \times 5$ grid in 21 seconds. The blackouts of $h + m$

points on the grids $5 \times 6$, $5 \times 7$, $6 \times 6$, $6 \times 7$, and $7 \times 7$ are stored in `jis/data/small_cells_sat_*.json`. The SAT solvers are CaDiCaL 1.5.3 and 1.9.5, Glucose 3, 4, and 4.2, MiniSat 2.2 [12], MapleChrono, and Lingeling from pysat 1.9.dev15 [20], and the algebra uses SymPy 1.14 [29].

A SAT instance asks for a valid blackout of at least $k$ points, so by the observation after Corollary 3 an unsatisfiable instance proves $f(h, m) < k$. Encoding A has a variable for each point and a clause "not all black" for each inclusion-minimal symmetric difference of two rectangles, with a sequential counter [36] for the size. Encoding B has a clause for each pair of rectangles that share a corner and an indicator for each hidden rectangle, at most one of them true, with a totalizer [4] for the size. The third encoding, `cells_sat_check.py`, is of type A but written from scratch. The structured encoders of Section 7 add the column structure of Theorem 11 for one row partition and keep only the clauses that involve a tilted rectangle. The collar test boards of Section 7 are checked by three separately written enumerators, which list every rectangle of the board and compare the presentations. The column searches call no solver. They place columns from left to right and stop when two rectangles inside the placed columns show the same points, which is sound since every interval of columns of a valid blackout is valid. The search `table_counts_search.py` also stops when the remaining columns cannot reach the target by the values of $f$ on narrower grids. So its counts on the $6 \times 7$ and $6 \times 8$ grids use $f(6, 6) \leq 12$, from the SAT runs and the ILP program, and those on the $6 \times 8$ and $7 \times 7$ grids use the SAT bound $f(6, 7) \leq 13$. The five-row and six-row searches use no such bound.

We did not produce DRAT or LRAT proofs. So $f(6, 7) < 14$, $f(7, 7) < 15$, $f(6, 8) < 16$, $f(7, 8) < 17$, and the ten row partitions that exclude $m + 9$ points at $m = 14$ in Theorem 43(a) rest on unsatisfiable SAT instances without a checked certificate, and on the agreement of independent encodings on different solvers. Every other upper bound also follows from a proof by hand, the ILP program, or an exhaustive search with no SAT solver. The ILP program gives no certificate either, so $f(6, 6) < 13$ rests on solvers alone. Found blackouts need no such trust, since the stored witnesses and every model of the counting programs are checked against the definition. A count needs in addition that the enumeration is complete.

The file `jis/scripts/README.md` gives, for each computer-assisted result, the command, the evidence file with its SHA-256 hash, and the stored time with what it measures. The file `requirements.txt` pins the package versions. The counts of the optima on the grids $5 \times 6$ to $7 \times 7$ in Table 2 come from two

SAT encodings that list every optimum, on Glucose 4 and MiniSat 2.2, and from a column search, all three in the scripts `table_counts_*`. On the $7 \times 8$ grid no valid blackout of 17 points exists, by two SAT runs split by the first column, so $f(7, 8) = 16$. One completed enumeration finds 18 optima there. The second count did not finish, so the table omits this count. Table 2 collects the values, and the rectangle counts of the programs agree with A085582 and A289832. Babichev and Babichev [3] count the rectangles of the square grid in near-linear time.

The Lean 4 files in `kagey-problems/lean/KageyProblems` check Lemma 2, Corollary 3, the fact that a valid blackout hides at most one rectangle, and Theorem 5. They also check the second half of Lemma 7, the part of Lemma 9 that the bound needs for one choice of trees, and Theorem 8, with Mantel's theorem from Mathlib. A finite list of rectangles is proved to hold exactly the rectangles of the grid. With it the Lean kernel decides that $S_0$, $S_8$, the blackout of Theorem 44(a), the stored blackout of the $6 \times 15$ grid, and the $5 \times 13$ and $6 \times 23$ test boards are valid. It also decides that the blackout $\{(1, 2), (2, 1), (3, 2), (2, 3)\}$ of the $4 \times 3$ grid hides $D(1, 1)$. So $f(5, 8) = 14$ and $f(6, 15) = 24$ are checked in full, and so are $f(5, 5) \geq 10$ and $f(6, 8) \geq 15$. No proof uses `sorry` or `native_decide`, so the Lean compiler is not trusted. Lemma 4, the collar theorem and so the values for every width, the rectangle counts, the generating functions, the bound $f(5, 5) \leq 10$, and the upper bounds from SAT runs are not formalized. Neither are the other results of Sections 5 to 8. Code and data are in the folder `kagey-problems/problem001` of the repository [34].

# 10 Three-row generating functions

We finish the proof of Theorem 25, with $D = LN = 1 - 2z - 4z^3 - 4z^4$. A *restricted run* is a string of singletons with no two letters $b$ at distance two. Reversal and reflection carry the next lemma to $BA$, $BC$ and $CB$.

**Lemma 51** (forbidden windows in the classes containing $B$)**.** *Let $S$ be a candidate in the ordered class $AB$ or in the triple class. A window fails if and only if the table lists it for its class, where $s$ is any singleton.*

| grid | $f$ | $a_h(m)$ | by |
|---|---|---|---|
| $2\times2$ | 4 | 1 | HIB |
| $2\times3$ | 4 | 12 | HIB |
| $2\times4$ | 5 | 32 | HIB |
| $2\times5$ | 6 | 80 | IB |
| $2\times6$ | 7 | 192 | TI |
| $2\times7$ | 8 | 448 | TI |
| $3\times3$ | 5 | 53 | IB |
| $3\times4$ | 6 | 224 | IB |

| grid | $f$ | $a_h(m)$ | by |
|---|---|---|---|
| $3\times5$ | 7 | 892 | IB |
| $3\times6$ | 8 | 3,420 | IB |
| $3\times7$ | 9 | 12,704 | IB |
| $3\times8$ | 10 | 45,864 | IB |
| $3\times9$ | 11 | 161,992 | IB |
| $3\times10$ | 12 | 562,912 | ITD |
| $4\times4$ | 7 | 316 | IB |
| $4\times5$ | 8 | 1,810 | IB |

| grid | $f$ | $a_h(m)$ | by |
|---|---|---|---|
| $4\times6$ | 10 | 26 | IB |
| $4\times7$ | 11 | 292 | IB |
| $4\times8$ | 12 | 2,118 | IMD |
| $5\times5$ | 10 | 8 | IB |
| $5\times6$ | 11 | 76 | ISC |
| $5\times7$ | 12 | 1,184 | ISC |
| $5\times8$ | 14 | 4 | TSDC |
| $5\times9$ | 15 | 44 | TSC |

| grid | $f$ | $a_h(m)$ | by |
|---|---|---|---|
| $5\times10$ | 16 | 350 | TSC |
| $5\times11$ | 17 | 3,004 | TSC |
| $5\times12$ | 18 | 24,706 | TSCM |
| $6\times6$ | 12 | 160 | ISC |
| $6\times7$ | 13 | 2,536 | SC |
| $6\times8$ | 15 | 8 | SC |
| $6\times15$ | 24 | – | TS |
| $7\times7$ | 14 | 7,300 | SC |
| $7\times8$ | 16 | – | S |

Table 2: Computed values for $h \le m$ and the methods that agree. H is by hand, T a theorem, I the ILP program (value only on $3\times10$, $4\times8$, $5\times6$, $5\times7$, and $6\times6$), B brute force, D every equality candidate, C a column search, S two SAT encodings, M a transfer counter, and a dash not computed. For the counts, S and C enumerate every optimum.

| *class* | *windows* | *forbidden* |
|---|---|---|
| *AB, X* | *singletons only* | *bab, bbb, bcb* |
| *AB* | *one double* | *bsA, Asb, bAb, bAc, cAb, aBa, aBb, bBa, cBc* |
| | *two doubles* | *bAB, ABa* |
| *X* | *one triple* | *bsX, Xsb, aXa, aXb, bXa, bXc, cXb, cXc* |

*Proof.* The tests (a) to (e) are those of Lemma 20, and Lemma 21(b) gives the singleton windows. A window $sAt$ fails exactly when it does in Lemma 22(a). Only the reason for $bAb$ changes, since $C \notin F$. Test (e) now fails it through $1 \in B_{i+1}$ and $B \in F$. In $sBt$ and $sXt$ both corners of the diamond in column $i+1$ are black. The diamond is empty if $s = t = b$. If $s = b \neq t$ it shows $(i+2, 2)$ alone and fails by test (e) exactly when $\{2, t\} \in F$, and symmetrically for $t = b \neq s$. If $s$ and $t$ are outer it shows the horizontal pair and fails by test (c) exactly when $s = t$. In $uAB$ the diamond fails by test (d) for $u = b$, since $(i+2, 3)$ and $(i+1, 2)$ are black, and shows three corners otherwise. In $ABw$ it is empty for $w = b$ and otherwise fails by test (e) exactly for $w = a$. In $Asb$ and $Xsb$ it shows the vertical pair or one point

of column $i+1$, and tests (b) and (e) fail it as $B \in F$. Every other window with a double or $X$ at the left tip shows three corners or an oblique pair whose test (d) needs the white point $(i+1, 2)$, so it passes. Reversal gives the right tip. □

We count restricted runs by a transfer on the last two bits. An outer letter has the bit 0 and the weight 2, and $b$ has the bit 1. Let $h_{ij}$ count, by $z^n$, the continuations of length $n \geq 0$ of a restricted run ending in the bits $i, j$. Since $b$ is forbidden after $1, j$, $h_{00} = 1 + z(2h_{00} + h_{01})$, $h_{01} = 1 + z(2h_{10} + h_{11})$, $h_{10} = 1 + 2zh_{00}$ and $h_{11} = 1 + 2zh_{10}$, and eliminating gives $Dh_{00} = 1 + z + 3z^2 + 2z^3$. Read runs $w_1 \cdots w_n$ with $n \geq 1$ outward from a double. Let $g_0$ count those with $w_1$ a prescribed outer letter and $g_1$ those with $w_1 = b$. Let $u$ and $v$ count the same runs with $w_2$, if present, outer. A fictitious outer letter before $w_1$ restricts nothing, so $g_0 = zh_{00}$, $g_1 = zh_{01}$, $u = z + 2z^2h_{00} = zh_{10}$ and $v = z + 2z^2h_{10} = zh_{11}$. Hence

$$Dg_0 = z(1+z+3z^2+2z^3), \;\; Dg_1 = z(1+z), \;\; Du = z(1+2z^2+2z^3), \;\; Dv = z. \tag{12}$$

**Proposition 52** (the triple class)**.** *In the notation of* (12)*,* $G_X(z) = z\big(1 + 4u + 2v + 2u^2 + v^2\big)$ *has a denominator dividing* $D^2$*.*

*Proof.* In $UXW$ the runs are restricted, and $bsX$ and $Xsb$ make the second letters from $X$ outer (Lemma 51). So one nonempty run gives $2u + v$ on either side. With two, the window at $X$ allows only the neighbor pairs $(a, c)$, $(b, b)$ and $(c, a)$, which give $2u^2 + v^2$. □

A word of $AB$ is $UAVBW$ with restricted runs $U$, $V$ and $W$. By Lemma 51, a nonempty $U$ read from $A$ has an outer second letter and is counted by $u$ or $v$, and a nonempty $W$ read from $B$ is counted by $g_0$ or $g_1$. Let $\ell_t$ and $r_t$ count the allowed $U$ and $W$, the empty one included, when $t$ stands right of $A$ or left of $B$. The windows centered at the double give

$$(\ell_a, \ell_b, \ell_c) = (1+2u+v, 1+u, 1+2u), \; (r_a, r_b, r_c) = (1+g_0, 1+g_0+g_1, 1+g_0+g_1). \tag{13}$$

For $s \in \{00, 10\}$ and a bit $\beta$ let $J_{s\beta}$ count, as for $h$, the continuations from $s$ after which the last bit is $\beta$. The empty continuation counts, with $\beta$ the second bit 0 of $s$. With rows $s = 00, 10$ and columns $\beta = 0, 1$, solving as for $h$ gives

$$J = \frac{1}{D}\begin{pmatrix} 1 + 2z^2 + 2z^3 & z + z^2 \\ 1 & 2z^2 + 2z^3 \end{pmatrix}. \tag{14}$$

**Proposition 53** (the class $\{A, B\}$, polynomial form computer-assisted)**.** *In the notation of* (12) *to* (14)*,*

$$\begin{aligned} G_{AB}(z) = z^2(1+2u)(1+g_0+g_1) + z^3(\ell_a r_a + \ell_b r_b + \ell_c r_c) \\ + z^4\,(\ell_a + \ell_c,\ \ell_b)\,J\,(r_a + r_c,\ 2r_b)^T. \end{aligned} \tag{15}$$

*Moreover* $G_{AB} = P_B/D^3$ *with*

$$P_B(z) = z^2(1+z)(1+z^2-3z^3+4z^6+4z^7). \tag{16}$$

*Proof.* The runs cover the windows inside them, and we split by the length $k$ of $V$. For $k = 0$ the windows $uAB$ and $ABw$ require $u \neq b$ and $w \neq a$, which gives the first term. For $k = 1$ the window $AtB$ passes, and $uAt$ and $tBw$ give $\ell_t$ and $r_t$. For $k \geq 2$ the window $Av_1v_2$ makes $v_2$ outer, $v_{k-1}v_kB$ passes, and the weight is $\ell_{v_1} r_{v_k}$. After $v_2$ the bits are 00 or 10 as $v_1$ is outer or $b$, and $v_3 \cdots v_k$ is a continuation from these bits. Summing over the last letter gives $J_{s0}(r_a + r_c) + J_{s1}\,2r_b$. The sum $r_a + r_c$ already counts both choices of an outer $v_k$. For $k \geq 3$ the weight 2 that $J$ gives an outer $v_k$ stands for the two choices of $v_2$, and so does the factor 2 in $2r_b$. For $k = 2$ the continuation is empty and $v_2 = v_k$, which is the constant term of $J$.

The script `three_row_closed_form.py` finds (16) in SymPy. □

**Theorem 54** (the exact denominator, computer-assisted)**.** *The reduced denominator of* $\sum_{m\geq 3} a_3(m)z^m$ *is* $Q = R^3L^3N^3$*, with five triple poles.*

*Proof.* By Theorem 25(a) the denominator divides $R^3D^3 = Q$. The factors $R$, $L$ and $N$ are coprime, since $L - N = -2z - 4z^2$ vanishes only at 0 and $-\frac{1}{2}$, where $N \neq 0$, and $L$ and $N$ are positive at $\frac{1}{3}$. The numerator in (2) is $\frac{4}{81}$ at $z = \frac{1}{3}$, where $G_{AB}$ and $G_X$ are analytic. Division gives $P_B \bmod L = \frac{z}{2}$ and $P_B \bmod N = \frac{1}{4} - \frac{z}{2}$, so $P_B$ vanishes at no root of $L$ or $N$. At these roots $G_{AB}$ has a pole of order three, $G_X$ one of order at most two, and $G_{AC}$ none. □

# 11 Square grids

This section proves the lower bound in question (4) of Section 12. Throughout, $n \geq 2$, the grid is $n \times n$, $\mathcal{R}$ is the set of its rectangles, tilted ones included, and $L = 1 + \ln n$ with the natural logarithm. For a point $v$ we write $d_v$ for the number of rectangles that have $v$ as a corner. By Lemma 4, two different rectangles share at most two corners, so the symmetric difference of their corner sets has four, six, or eight points.

## 11.1 Pairs with two common corners

**Lemma 55** (two common corners)**.** *Let $R_1 \neq R_2$ be rectangles with exactly two common corners $p$ and $q$. If $\{p, q\}$ is a side of both rectangles, or a diagonal of both, then $C(R_1) \triangle C(R_2)$ is the corner set of a rectangle.*

*Proof.* Suppose $\{p, q\}$ is a side of both. The other corners are $p+u, q+u$ for $R_1$ and $p+v, q+v$ for $R_2$, where $u \neq v$ are nonzero vectors perpendicular to $q-p$. The four points $p+u$, $q+u$, $p+v$, $q+v$ are distinct, since $p+u = q+v$ would make $v - u = p - q$ a nonzero vector perpendicular to itself. None of them is $p$ or $q$, since the common corners are exactly $p$ and $q$. They form a parallelogram with sides $q-p$ and $v-u$, which are nonzero and perpendicular, so they are the corners of a rectangle.

Suppose $\{p, q\}$ is a diagonal of both. Both rectangles have center $o = (p + q)/2$, and their other corners are $o \pm u$ for $R_1$ and $o \pm v$ for $R_2$ with $|u| = |v| = |p - q|/2$ and $v \neq \pm u$. The segments from $o - u$ to $o + u$ and from $o - v$ to $o + v$ have the same midpoint and the same length. So $o \pm u$ and $o \pm v$ are the four distinct corners of a rectangle. □

## 11.2 Lattice points on circles

Let $r_2(t)$ be the number of pairs $(a, b)$ of integers with $a^2 + b^2 = t$, and let $\tau(t)$ be the number of positive divisors of $t$.

**Lemma 56** (divisor bound)**.** *For every $t \geq 1$, $r_2(t) \leq 4\tau(t)$. For every $\varepsilon > 0$ there is a constant $K_\varepsilon$ such that $\tau(t) \leq K_\varepsilon t^\varepsilon$ for all $t \geq 1$.*

*Proof.* Both parts are classical. Hardy and Wright [19, Sections 16.9 and 16.10] prove that $r_2(t) = 4\big(d_1(t) - d_3(t)\big)$, where $d_i(t)$ is the number of divisors of $t$ congruent to $i$ modulo 4, so $r_2(t) \leq 4d_1(t) \leq 4\tau(t)$. The bound $\tau(t) = O(t^\varepsilon)$ for every fixed $\varepsilon > 0$ is proved in their Section 18.1. □

## 11.3 Counting rectangles and obstructions

Let $\mathcal{A}$ be the family of four-point sets consisting of all corner sets $C(R)$ with $R \in \mathcal{R}$, together with the mixed differences. A *mixed difference* is a set $C(R_1) \triangle C(R_2)$ for a pair $R_1 \neq R_2$ with exactly two common corners that form a side of one rectangle and a diagonal of the other, the case that Lemma 55 leaves out. Let $\mathcal{B}$ be the family of all six-point sets $C(R_1) \triangle C(R_2)$ for pairs $R_1 \neq R_2$ with exactly one common corner.

**Lemma 57** (rectangle and obstruction counts)**.** *Let $n \geq 2$ and $L = 1 + \ln n$.*

*(a) For every point $v$, $d_v \leq 8n^2L$, and $|\mathcal{R}| \leq 2n^4L$.*

*(b) For every $\varepsilon > 0$, $|\mathcal{A}| \leq A_\varepsilon n^{4+2\varepsilon}L$ with $A_\varepsilon = 2(1 + 16 \cdot 2^\varepsilon K_\varepsilon)$, and $|\mathcal{B}| \leq 32n^6L^2$.*

*(c) If a blackout $S$ of the $n \times n$ grid contains no member of $\mathcal{A}$ and no member of $\mathcal{B}$, then $S$ is valid.*

*Proof.* (a) Let $R$ be a rectangle with corner $v$. Its two sides from $v$ are perpendicular nonzero integer vectors. Write the first as $k(a, b)$ with $k \geq 1$ and $\gcd(a, b) = 1$. The second is perpendicular to $(a, b)$, so it is $(-bl, al)$ for a real $l$, and $l$ is a nonzero integer because the vector is integral and $\gcd(a, b) = 1$. Put $d = \max(|a|, |b|) \geq 1$. There are $8d$ integer vectors $(a, b)$ with $\max(|a|, |b|) = d$, since they lie on the boundary of the square $[-d, d]^2$. Both sides fit in the grid, so $kd \leq n - 1$ and $|l|d \leq n - 1$, which leaves at most $(n - 1)/d$ values of $k$ and $2(n - 1)/d$ values of $l$. The ordered pair of sides determines $k$, $(a, b)$ and $l$, and $R$ gives two ordered pairs, so

$$2d_v \leq \sum_{d=1}^{n-1} 8d \cdot \frac{n-1}{d} \cdot \frac{2(n-1)}{d} \leq 16n^2 \sum_{d=1}^{n-1} \frac{1}{d} \leq 16n^2L.$$

Each rectangle has four corners, so $4|\mathcal{R}| = \sum_v d_v \leq n^2 \cdot 8n^2L$.

(b) Fix $R_1 \in \mathcal{R}$ and one of its four sides $\{p, q\}$. A rectangle $R_2$ that has $\{p, q\}$ as a diagonal has center $(p + q)/2$ and a second diagonal of the same length, joining two grid points $p', q'$. The vector $p' - q'$ is an integer vector with $|p' - q'|^2 = |p - q|^2 = t$, where $t \leq 2(n - 1)^2 < 2n^2$, and it determines $p'$ and $q'$. So there are at most $r_2(t) \leq 4K_\varepsilon(2n^2)^\varepsilon$ such $R_2$ by Lemma 56. Every mixed difference arises from some $R_1$, side, and $R_2$ in this way, so

$$|\mathcal{A}| \leq |\mathcal{R}|\big(1 + 4 \cdot 4K_\varepsilon 2^\varepsilon n^{2\varepsilon}\big) \leq 2n^4L\big(1 + 16 \cdot 2^\varepsilon K_\varepsilon\big)n^{2\varepsilon} = A_\varepsilon n^{4+2\varepsilon}L.$$

A member of $\mathcal{B}$ comes from an unordered pair of rectangles with a common corner $v$, so

$$|\mathcal{B}| \leq \sum_v \binom{d_v}{2} \leq \frac{n^2}{2}\big(8n^2L\big)^2 = 32n^6L^2.$$

(c) Suppose $S$ is not valid, so some $R_1 \neq R_2$ have the same presentation, and $C(R_1) \triangle C(R_2) \subseteq S$ by Lemma 2. If $R_1$ and $R_2$ have two common corners,

the symmetric difference is the corner set of a rectangle by Lemma 55 or a mixed difference, and in both cases it is a member of $\mathcal{A}$. If they have one common corner, it is a member of $\mathcal{B}$. If they have none, then $C(R_1) \subseteq S$ is a member of $\mathcal{A}$. In each case $S$ contains a member of $\mathcal{A} \cup \mathcal{B}$. □

Avoiding $\mathcal{A}$ forbids the one fully black rectangle that the game allows, which is harmless for a lower bound.

## 11.4 The lower bound

**Theorem 58** (random blackouts with deletions)**.** *We have*

$$\liminf_{n\to\infty} \frac{f(n,n)\,(1+\ln n)^{2/5}}{n^{6/5}} \geq \frac{5}{6\cdot 192^{1/5}} = 0.2911\ldots,$$

*so for every $c < 5/(6\cdot 192^{1/5})$ we have $f(n,n) \geq c\,n^{6/5}(1+\ln n)^{-2/5}$ for all large $n$. In particular $f(n,n) - 2n \to \infty$.*

*Proof.* Put $a = 192^{-1/5}$, $T = n^{6/5}L^{-2/5}$ and $\theta = a\,n^{-4/5}L^{-2/5}$, which lies in $(0,1)$ since $a < 1$, and select each point of the grid independently with probability $\theta$. Let $X$ be the number of selected points. Let $Y$ be the number of members of $\mathcal{A}$ all of whose four points are selected, and $Z$ the number of members of $\mathcal{B}$ all of whose six points are selected. By linearity of expectation,

$$\mathbb{E}(X - Y - Z) = n^2\theta - |\mathcal{A}|\theta^4 - |\mathcal{B}|\theta^6. \tag{17}$$

Here $n^2\theta = aT$. With $\varepsilon = 1/10$ and $A_0 = A_{1/10}$, Lemma 57(b) gives $|\mathcal{B}|\theta^6 \leq 32n^6L^2a^6n^{-24/5}L^{-12/5} = 32a^6T$ and $|\mathcal{A}|\theta^4 \leq A_0 n^{21/5}La^4n^{-16/5}L^{-8/5} \leq A_0a^4n^{-1/5}T$, as $L \geq 1$. Since $32a^5 = \frac{1}{6}$, (17) gives $\mathbb{E}(X-Y-Z) \geq \left(\frac{5}{6}a - A_0a^4n^{-1/5}\right)T$, so some selection $S_0$ has $X - Y - Z$ at least this value.

Fix this $S_0$. Go through the members of $\mathcal{A}\cup\mathcal{B}$ that lie inside $S_0$, and for each one that is still contained in the current set delete one of its points. This deletes at most $Y + Z$ points and leaves a set $S \subseteq S_0$ with $|S| \geq X - Y - Z$. No member of $\mathcal{A} \cup \mathcal{B}$ is contained in $S$. Indeed a member contained in $S$ is contained in $S_0$, so it was visited, and at that moment it lost a point or had already lost one, and the families do not depend on the deletions. By Lemma 57(c), $S$ is a valid blackout, so $f(n,n)/T \geq \frac{5}{6}a - A_0a^4n^{-1/5}$, which tends to $\frac{5}{6}a = 5/(6\cdot 192^{1/5})$. Finally $T/n \to \infty$ gives the last claim. □

The value $a = 192^{-1/5}$ maximizes $a - 32a^6$, and every $0 < \varepsilon < 1/5$ would do in place of $1/10$. This is the alteration method of Alon and Spencer [2, Section 3.2].

## 12 Open questions

(1) *Five-row counts.* The series $\sum_{m\ge 0} a_5(m) z^m$ is rational by Theorem 49. Find it, and the growth rate of $a_5(m)$ (Remark 42).

(2) *Six rows below width* 15. The first stored blackout in `jis/data/small_cells_sat_6x9_b16.json`, found by encoding A on CaDiCaL, passes the test of Theorem 39 with seven columns black only in row 1 added on each side (`six_row_collar_6x9_check.py`). With Theorem 43 this gives $m+7 \le f(6,m) \le m+8$ for $9 \le m \le 14$. What are these values?

(3) *Seven and more rows.* Is the row bound $m + q_h$ attained at every large width for every $h \ge 7$? By Theorem 39, one valid $7 \times m$ blackout of $m+12$ points that passes the collar test would settle $h = 7$.

(4) *Squares.* For $n \ge 3$, Lemma 7 and Neuwirth's bound for bipartite graphs of girth eight [32, Corollary 16] give the known bound $f(n,n) < n^{4/3} + \frac{2}{3}n - \frac{2}{9}n^{2/3} - \frac{20}{81}n^{1/3}$. Theorem 58 gives $f(n,n) \ge 0.29\, n^{6/5}(1+\ln n)^{-2/5}$ for large $n$, and Lavrov [22] used the same method for the triangular version. What is the true order, and the least $n$ with $f(n,n) > 2n$?

(5) *Proofs by hand.* Is there a proof without a computer that the $5 \times 5$ optimum is unique up to symmetry, or that the four-row generating function has a dominant pole of order five?

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